Rockchip USB SQ Test Guide

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Preface

Overview

This document provides methods for USB 2.0/3.0 signal integrity (SQ) testing on the Rockchip platform.

USB 2.0 signal integrity testing includes High Speed, Full Speed, and Low Speed modes. Test items include: High Speed Signal Quality, Packet Parameters, CHIRP Timing, Suspend/Resume/Reset Timing, Test J/K, SE0_NAK, Receiver Sensitivity, etc. This document provides the commonly used test methods for High Speed Signal Quality and Receiver Sensitivity.

USB 3.0 signal integrity testing includes Tx compliance test and Rx compliance test. Since the Rx compliance test environment and methods are relatively complex, this document does not provide detailed Rx test methods, but only provides detailed Tx test methods and an explanation of the Rx test principle.

Rockchip SOCs typically integrate multiple USB controllers that are independent of each other. For detailed information, please refer to the corresponding chip TRM. Different USB controllers use different test commands and test methods. Therefore, before testing USB signal integrity, please first identify the USB controller corresponding to the USB interface to be tested.

The test methods provided in this document are applicable to Agilent, Tektronix, and LeCroy oscilloscopes and USB test fixtures.

The test commands provided in this document are explained based on the USB controller model used by the USB interface, which requires a certain level of software technical background. If testers wish to directly find the corresponding test commands through the USB interface names in the hardware schematic, please use Rockchip USB SQ Tool.

Product Version

Chip Models	Kernel Version
All Rockchip chips	All kernel versions

Intended Audience

This document (this guide) is primarily intended for the following engineers:

- Hardware Engineers
- Software Engineers
- Technical Support Engineers

Revision History

Version	Author	Date	Description
V1.0	Wu Liangfeng	2017-12-12	Initial version
V1.1	Wu Liangfeng	2018-03-07	Added rk3399 Type-C reverse side test commands
V1.2	Wu Liangfeng	2018-03-30	Added RK3229/RK3326/PX30 test commands
V1.3	Wu Liangfeng	2018-05-21	Corrected USB 3.0 test method
V1.4	Wu Liangfeng	2019-01-09	Revised format using markdownlint
V1.5	Wu Liangfeng	2019-01-15	Added RK1808 test commands, corrected document indentation, corrected USB 3.0 test commands
V1.6	Wu Liangfeng	2019-06-26	Added solutions for failure to enter Compliance Test
V1.6.1	Wu Liangfeng	2019-10-21	Added RK2108/RK2206 test commands
V1.6.2	Wu Liangfeng	2020-02-19	Added disclaimer, trademark statement, and copyright statement
V1.6.3	Ren Jianing	2020-12-08	Added RK3566/RK3568 test commands
V1.7.0	Wang Mingcheng	2020-12-16	Added RK625 test commands
V1.8.0	Wu Liangfeng	2021-12-22	Added RK3588/RK3588S/RK3308B-S/RK3308H-S/PX30S/RK3326S test commands, added RK3588 USB 3.1 test notes
V1.9.0	Wu Liangfeng	2022-04-16	Added RV1103/RV1106 test commands; added USB SQ Tool description and download link
V2.0.0	Wu Liangfeng	2022-06-14	Added USB 2.0 Rx test description, updated USB 3.0 test description, updated USB HUB test description, revised table and image formats
V2.1.0	Wu Liangfeng	2022-12-04	Added RK3528 test commands
V2.2.0	Wu Liangfeng	2023-01-15	Added RK3562 test commands
V2.3.0	Wu Liangfeng	2024-04-23	Added RK3576/RK2118 test commands

Version	Author	Date	Description
V2.4.0	Wu Liangfeng	2024-08-06	Added RK3506/RV1103B test commands
V2.5.0	Wu Liangfeng	2025-04-25	Added RV1126B test commands
V2.6.0	Wu Liangfeng	2025-07-30	Added RK1820/RK2116 test commands
V2.7.0	Zheng Jianwei	2025-12-15	Added RK3538 test commands
V2.8.0	Wu Liangfeng	2026-04-09	Added RK3572 test commands
V2.9.0	Wu Liangfeng	2026-06-22	Corrected typos and technical errors

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1. USB 2.0 Compliance Test

1.1 USB 2.0 Compliance Test Content

USB 2.0 Tx Signal Quality test items

- Eye diagram test
- Signal rate
- Packet trailer width
- Crossover voltage range (for Low Speed and Full Speed)
- JK jitter, KJ jitter
- Continuous jitter
- Monotonicity test (for High Speed)
- Rise/fall time

USB 2.0 Rx Receiver Sensitivity test items

- Receiver sensitivity
- Noise threshold
- Minimum SYNC Field

1.2 USB 2.0 Tx Test Commands and Test Tools

The principle of USB 2.0 Tx SQ testing is to set the USB controller's Test Control register to make the USB controller enter Test Packet Mode, and the USB controller will continuously generate and send periodic Test Patterns. The USB oscilloscope analyzes the USB signal integrity by detecting the waveform of the Test Pattern.

For USB 2.0 Device and USB 2.0 Host interfaces on the Rockchip platform, the methods to set the USB controller to enter Test Packet Mode are different:

- For USB 2.0 Device, you can use test commands or test tools to set the USB controller to enter Test Packet Mode
- For USB 2.0 Host, you can only use test commands to set the USB controller to enter Test Packet Mode

1.2.1 USB 2.0 Device Tx Test Commands and Test Tools

USB 2.0 Device SQ Test Commands

The test commands are shown in Table 1 below. Commands can be executed via serial console or ADB.

Table 1 USB 2.0 Device SQ Test Commands

Chip Name	DWC2_0 OTG 2.0 Device	DWC3_0 OTG 2.0 device	DWC3_1 OTG 2.0 device
RK1808	N.A	io -4 0xfd00c704 0x8c000a08	N.A
RK1820	N.A	io -4 0x102500c704 0x8c000a08	io -4 0x102400c704 0x8c000a08
RK2108	io -4 0x41300804 0x40	N.A	N.A
RK2116	io -4 0x40040804 0x40	N.A	N.A
RK2118	io -4 0x50040804 0x40	N.A	N.A
RK2206	io -4 0x43040804 0x40	N.A	N.A
RK29XX RK30XX RK31XX	io -4 0x10180804 0x40	N.A	N.A
RK3228 RK3229	io -4 0x30040804 0x40	N.A	N.A
RK3288 RK3228H RK3328 RK3368	io -4 0xff580804 0x40	N.A	N.A
RK3308 RK3308B-S RK3308H-S	io -4 0xff400804 0x40	N.A	N.A
RK3326 RK3326S PX30 PX30S	io -4 0xff300804 0x40	N.A	N.A
RV1103 RV1106	N.A	io -4 0xffb0c704 0x8c000a08	N.A
RV1103B	N.A	io -4 0x20b0c704 0x8c000a08	N.A
RV1108	io -4 0x30180804 0x40	N.A	N.A
RV1126 RV1109	N.A	io -4 0xffd0c704 0x8c000a08	N.A
RV1126B	N.A	io -4 0x2150c704 0x8c000a08	N.A
SOFIA-3GR	io -4 0xe2100804 0x40	N.A	N.A

Chip Name	DWC2_0 OTG 2.0 Device	DWC3_0 OTG 2.0 device	DWC3_1 OTG 2.0 device
RK3366	io -4 0xff4c0804 0x40	io -4 0xff50c704 0x8c000a08	N.A
RK3399	N.A	io -4 0xfe80c704 0x8c000a08	io -4 0xfe90c704 0x8c000a08
RK3506	io -4 0xff740804 0x40	N.A	N.A
RK3528	N.A	io -4 0xfe50c704 0x8c000a08	N.A
RK3538	N.A	io -4 0xfc00c704 0x8c000a08	N.A
RK3562	N.A	io -4 0xfe50c704 0x8c000a08	N.A
RK3572	N.A	io -4 0x2400c704 0x8c000a08	io -4 0x2440c704 0x8c000a08
RK3576	N.A	io -4 0x2300c704 0x8c000a08	io -4 0x2340c704 0x8c000a08
RK3566 RK3568	N.A	io -4 0xfcc0c704 0x8c000a08	N.A
RK625	io -4 0x40180804 0x40	N.A	N.A
RK3588 RK3588S	N.A	io -4 0xfc00c704 0x8c000a08	io -4 0xfc40c704 0x8c000a08

Note:

RK3506 OTG1 Device test command: io -4 0xff780804 0x40

USB 2.0 Device Test Tools

For USB 2.0 Device SQ testing on the Rockchip platform, in addition to the above test commands, you can also use the USB HSET test tool provided by the official USB-IF organization. Download the tool "USBHSET for EHCI" or "USBHSET for XHCI".

Download links are as follows:

USB 2.0 HOST (EHCI):

32 bit: https://www.usb.org/document-library/usbhset-ehci-32-bit

64 bit: https://www.usb.org/document-library/usbhset-ehci-64-bit

USB 3.0 HOST (xHCI):

32 bit: https://www.usb.org/document-library/xhsett-x32

64 bit: https://www.usb.org/document-library/xhsett-x64e

Brief usage steps of the test tool are as follows:

1. Connect the USB device port to be tested to the PC via a USB cable
2. Open the test tool, select "Device", and then click the "TEST" button, as shown in Figure 1 below

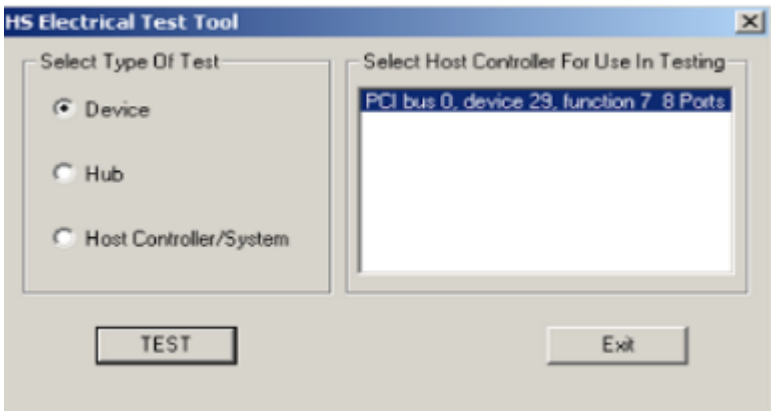


Figure 1 Select Test Type

3. As shown in Figure 2, select the device to be tested and the test command "TEST PACKET", then click the "EXECUTE" button. After completing the above operations, the USB controller will automatically enter Test Packet Mode and continuously send periodic test packets. The test waveform is shown in Figure 3:

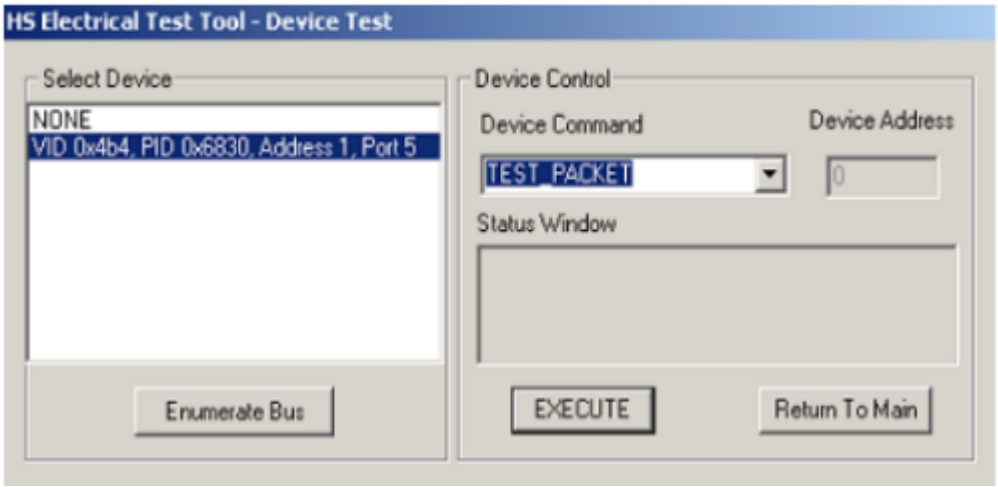


Figure 2 Select Test Device and Test Command

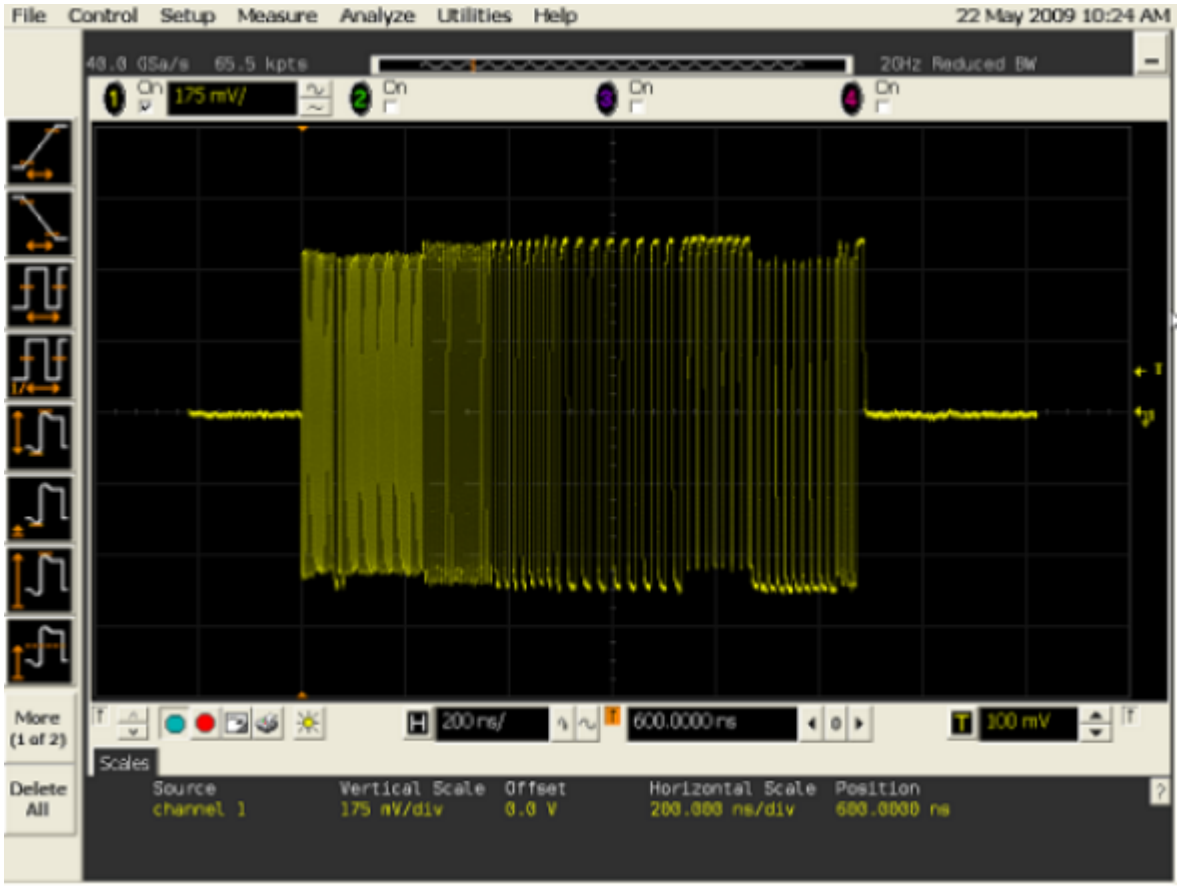


Figure 3 Test Waveform

1.2.2 USB 2.0 Host Tx Test Commands

USB 2.0 Host SQ testing can only use test commands; there is no dedicated test tool. The test commands are shown in Table 2, Table 3, and Table 4. The test commands can be executed via ADB or serial console.

Table 2 USB 2.0 Host SQ Test Commands (a)

Chip Name	DWC2_0 OTG Host 2.0	DWC2_1 Host 2.0	EHCI_0 Host 2.0
RK1808	N.A	N.A	io -4 0xffd80054 0x40000
RK2116	io -4 0x40040440 0x8000	N.A	N.A
RK2118	io -4 0x50040440 0x8000	N.A	N.A
RK2206	io -4 0x43040440 0x8000	N.A	N.A
RK29XX RK30XX RK3188	io -4 0x10180440 0x8000	io -4 0x101c0440 0x8000	N.A
RK312X	io -4 0x10180440 0x8000	io -4 0x101c0440 0x8000	io -4 0x101c0054 0x40000
RK3228 RK3229	io -4 0x30040440 0x8000	N.A	io -4 0x30080054 0x40000
RK3288	io -4 0xff580440 0x8000	io -4 0xff540440 0x8000	io -4 0xff500054 0x40000
RK3308 RK3308B-S RK3308H-S	io -4 0xff400440 0x8000	N.A	io -4 0xff440054 0x40000
RK3326 RK3326S	io -4 0xff300440 0x8000	N.A	N.A
PX30 PX30S	io -4 0xff300440 0x8000	N.A	io -4 0xff340054 0x40000
RK3228H RK3328	io -4 0xff580440 0x8000	N.A	io -4 0xff5c0054 0x40000
RK3366	io -4 0xff4c0440 0x8000	N.A	io -4 0xff480054 0x40000
RK3368	io -4 0xff580440 0x8000	N.A	io -4 0xff500054 0x40000
RV1108	io -4 0x30180440 0x8000	N.A	io -4 0x30140054 0x40000
RV1126 RV1109	N.A	N.A	io -4 0xffe00054 0x40000
RV1126B	N.A	N.A	io -4 0x21480054 0x40000
SOFIA-3GR	io -4 0xe2100440 0x8000	N.A	N.A
RK3399	N.A	N.A	io -4 0xfe380054 0x40000
RK3506	io -4 0xff740440 0x8000	io -4 0xff780440 0x8000	N.A
RK3528	N.A	N.A	io -4 0xff100054 0x40000
RK3538	N.A	N.A	io -4 0xfc100054 0x40000

Chip Name	DWC2_0 OTG Host 2.0	DWC2_1 Host 2.0	EHCI_0 Host 2.0
RK3562	N.A	N.A	io -4 0xfed00054 0x40000
RK3566 RK3568	N.A	N.A	io -4 0xfd800054 0x40000
RK3588 RK3588S	N.A	N.A	io -4 0xfc800054 0x40000

Table 3 USB 2.0 Host SQ Test Commands (b)

Chip Name	EHCI_1 Host 2.0	EHCI_2 Host 2.0	EHCI_HSIC Host 2.0
RK3168 RK3188	N.A	N.A	io -4 0x10240054 0x40000
RK3288	N.A	N.A	io -4 0xff5c0054 0x40000
RK3228 RK3229	io -4 0x300c0054 0x40000	io -4 0x30100054 0x40000	N.A
RK3368	N.A	N.A	io -4 0xff5c0054 0x40000
RK3399	io -4 0xfe3c0054 0x40000	N.A	io -4 0xfe340054 0x40000
RK3566 RK3568	io -4 0xfd880054 0x40000	N.A	N.A
RK3588 RK3588S	io -4 0xfc880054 0x40000	N.A	N.A

Table 4 USB 2.0 Host SQ Test Commands (c)

Chip Name	DWC3_0 OTG Host 2.0	DWC3_1 OTG Host 2.0
RK1808	io -4 0xfd000424 0x40000000	N.A
RK1820	io -4 0x1025000424 0x40000000	io -4 0x1024000424 0x40000000
RK3228H	io -4 0xff600424 0x40000000	N.A
RK3366	io -4 0xff500424 0x40000000	N.A
RK3399	io -4 0xfe800424 0x40000000	io -4 0xfe900424 0x40000000
RV1103 RV1106	io -4 0xffb00424 0x40000000	N.A
RV1103B	io -4 0x20b00424 0x40000000	N.A
RV1126 RV1109	io -4 0xffd00424 0x40000000	N.A
RV1126B	io -4 0x21500424 0x40000000	N.A
RK3528	io -4 0xfe500424 0x40000000	N.A
RK3538	io -4 0xfc000424 0x40000000	N.A
RK3562	io -4 0xfe500424 0x40000000	N.A
RK3566 RK3568	io -4 0xfcc00424 0x40000000	io -4 0xfd000424 0x40000000
RK3572	io -4 0x24000424 0x40000000	io -4 0x24400424 0x40000000
RK3576	io -4 0x23000424 0x40000000	io -4 0x23400424 0x40000000
RK3588 RK3588S	io -4 0xfc000424 0x40000000	io -4 0xfc400424 0x40000000

1.3 USB 2.0 Tx Test Environment

Testing USB 2.0 SQ requires an oscilloscope equipped with professional USB test software, such as Agilent 9000 series oscilloscope (or 90000 series), Tektronix, or LeCroy series oscilloscopes. This document mainly introduces the USB 2.0 test environment for the Agilent 9000 series oscilloscope. The required tools are as follows:

- MSO9254A oscilloscope with USB 2.0 test software N5416A installed
- 113xA differential active probe
- E2678A differential probe head
- E2649-66401 device fixture and E2649-66402 host fixture
- USB 2.0 cable

The USB 2.0 cable must strictly comply with the USB 2.0 Spec. If the USB cable length is less than 10cm, the Near End template should be used for eye diagram testing. If the cable length is greater than 10cm, the Far End template should be used for eye diagram testing. During Near End testing, using a USB cable that is too long or has poor impedance matching may cause eye diagram test metrics to fail.

1.4 USB 2.0 Tx Test Steps

1. Set up the test environment

If you are using the Agilent test suite, please refer to the following documents for test environment setup and oscilloscope configuration:

Agilent N5416A USB 2.0 Compliance Test Option

Agilent USB 2.0 High Speed Device SQ Test

If you are using Tektronix or LeCroy test suites, please search for the test instruction documents on the Tektronix and LeCroy official websites.

Taking the Agilent oscilloscope for USB 2.0 Device SQ testing as an example, the completed test environment is shown in Figure 4:

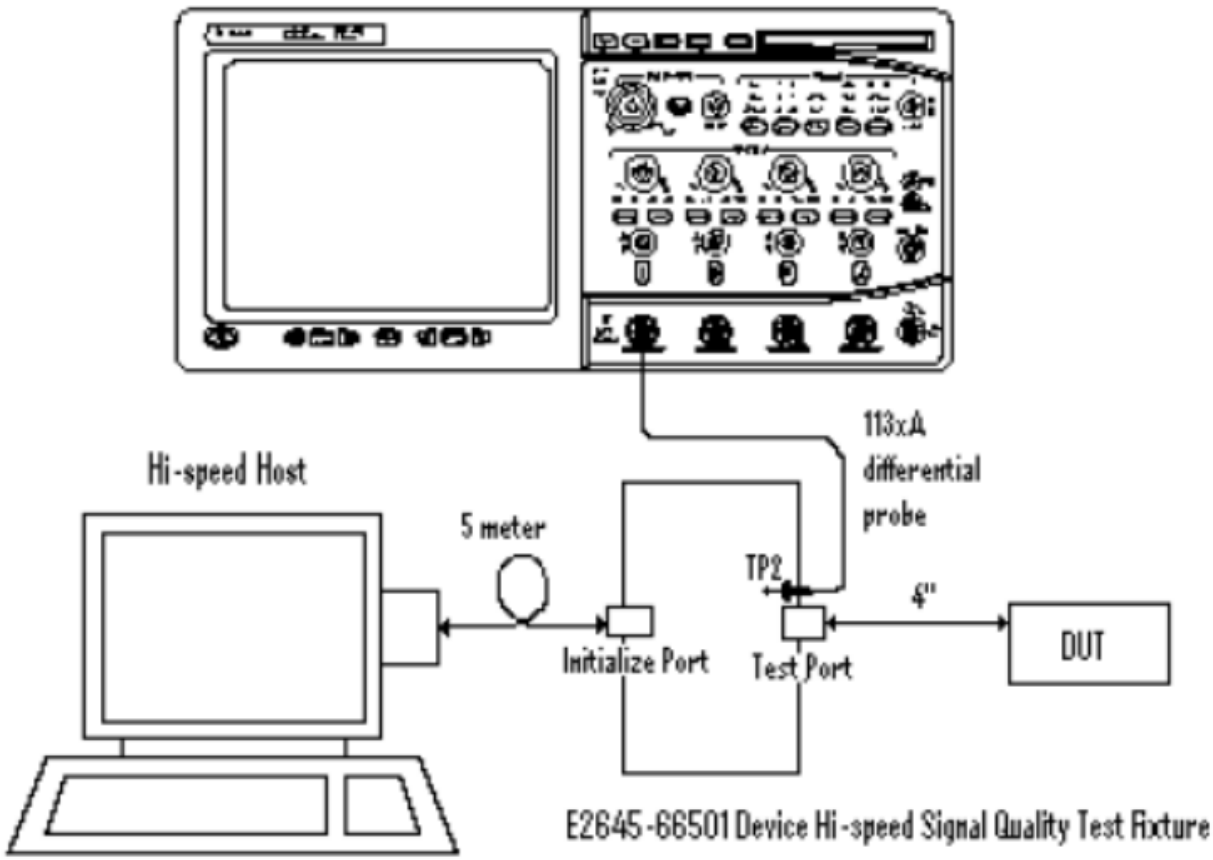


Figure 4 Agilent USB 2.0 SQ Test Environment

2. Set USB to enter test mode [Test Packet Mode]

Before setting the USB controller to enter Test Mode, you must first confirm that the USB can communicate normally.

If testing the USB 2.0 OTG Device interface, ensure that the USB port to be tested is connected to the PC through the test fixture and cables, and the PC can properly recognize the USB device. Then, refer to the PC-side test tool or test commands mentioned in the section "USB 2.0 Device Tx Test Commands and Test Tools" to set the USB controller to enter test mode.

If testing the USB 2.0 Host interface, the method to set the USB controller to enter test mode differs depending on the oscilloscope and test fixture. The following briefly describes the setup methods for using the Agilent test suite and the Tektronix test suite:

2.1 Agilent Test Suite

First, connect the Host port to be tested to one end of the test fixture's test interface, then connect a high-speed device (such as a USB flash drive) to the other end of the fixture, as shown in Figure 5 below. Then execute the test command to set the USB to test mode.

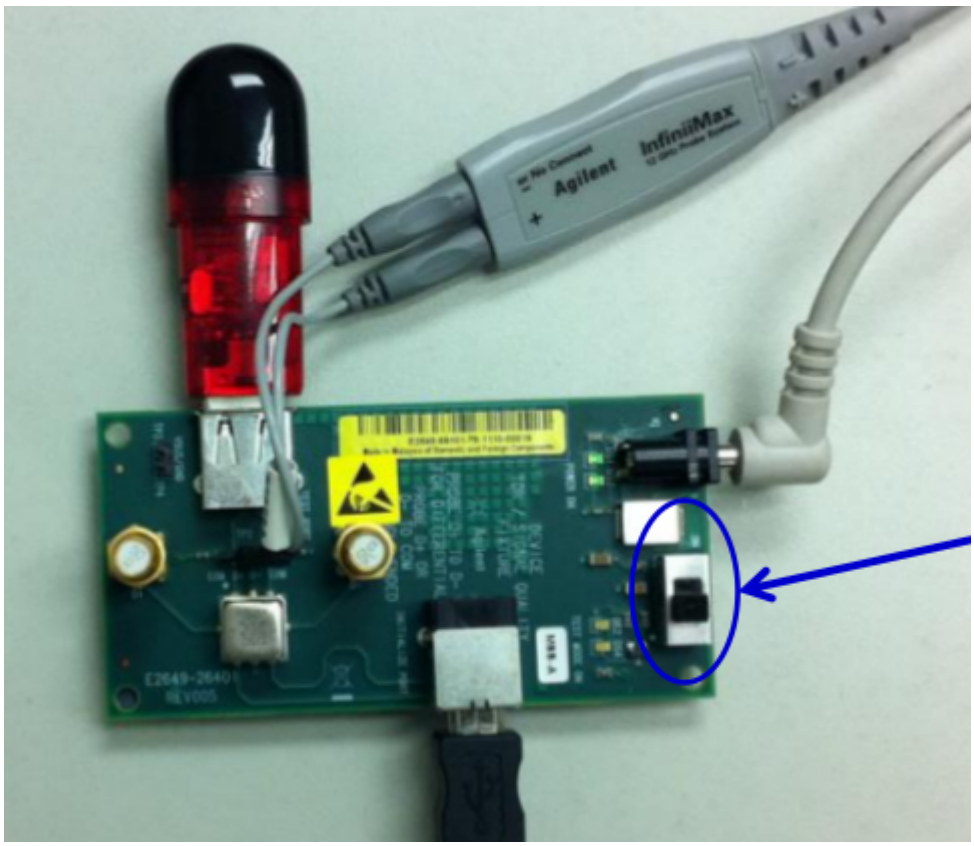


Figure 5 Agilent USB 2.0 SQ Test Fixture Connection Method

2.2 Tektronix Test Suite

If using the Tektronix test fixture, since there is only one interface on the test fixture for connecting the HOST port to be tested, and no other high-speed devices can be connected, first disconnect the Host port to be tested from the test fixture, then plug a high-speed USB device (such as a USB flash drive) into the Host port to be tested, and then execute the test command. After the test command is sent, the USB controller will automatically enter test mode. Then unplug the high-speed USB device, connect the test fixture to the Host port to be tested, and start testing.

*Note: When testing the USB 2.0 Host interface, a **high-speed USB device** (such as a USB flash drive) must be connected to the Host port. Do not connect full-speed or low-speed USB devices such as mice or keyboards.*

3. USB Automated Test Software Analyzes the Waveform

After the USB controller successfully enters test mode, it will generate and send continuous Test Packet Patterns, and periodic test waveforms can be observed from the oscilloscope, as shown in Figure 6. The USB automated test software on the oscilloscope will automatically analyze the test waveform and generate a complete test report.

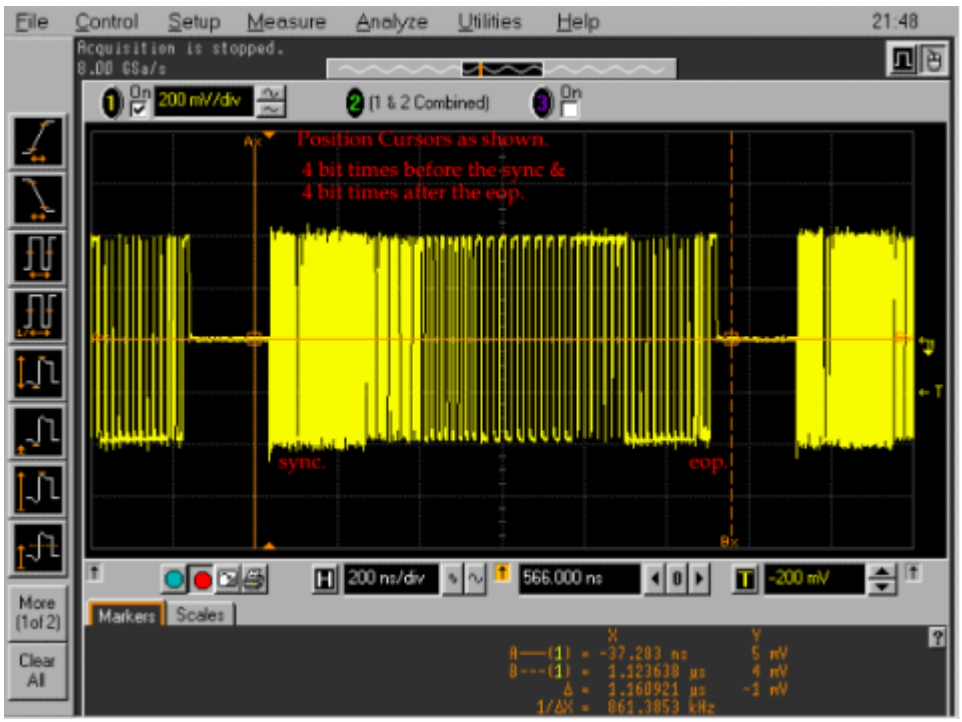


Figure 6 USB 2.0 Signal Quality Analysis Interface

1.5 USB 2.0 Eye Diagram Analysis

1.5.1 USB 2.0 Standard Eye Diagram Analysis

There are two different standards for USB 2.0 eye diagram templates: Near End and Far End. In High Speed Signal Quality testing, if the USB port under test is directly connected to the test fixture via a cable shorter than 10cm, the Near End eye diagram template is used. If the USB port under test is connected to the test fixture via a cable longer than 10cm, the Far End eye diagram template is used. In USB 2.0 eye diagram testing on the Rockchip platform, to ensure the reliability of USB 2.0 signal quality, it is recommended to uniformly adopt the stricter Near End eye diagram template as the reference standard. Figures 7 and 8 show the standard USB eye diagrams using the Near End and Far End eye diagram templates, respectively.

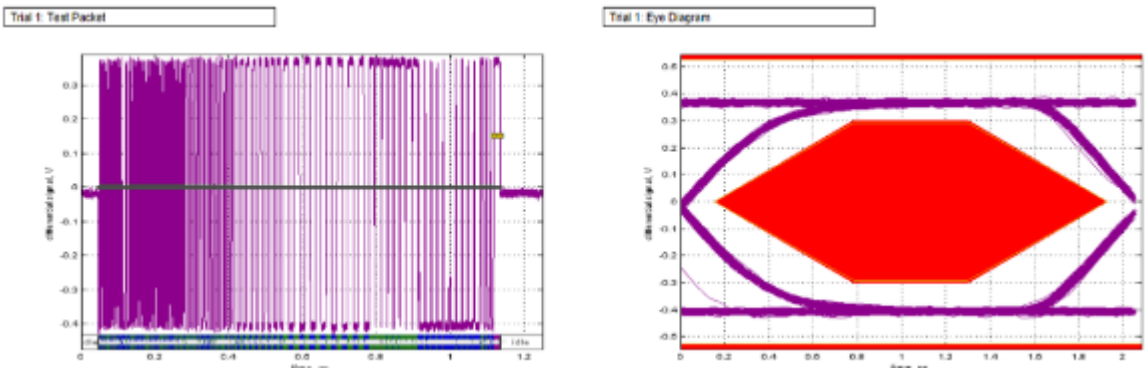


Figure 7 USB 2.0 High-speed Near End SQ Eye Diagram

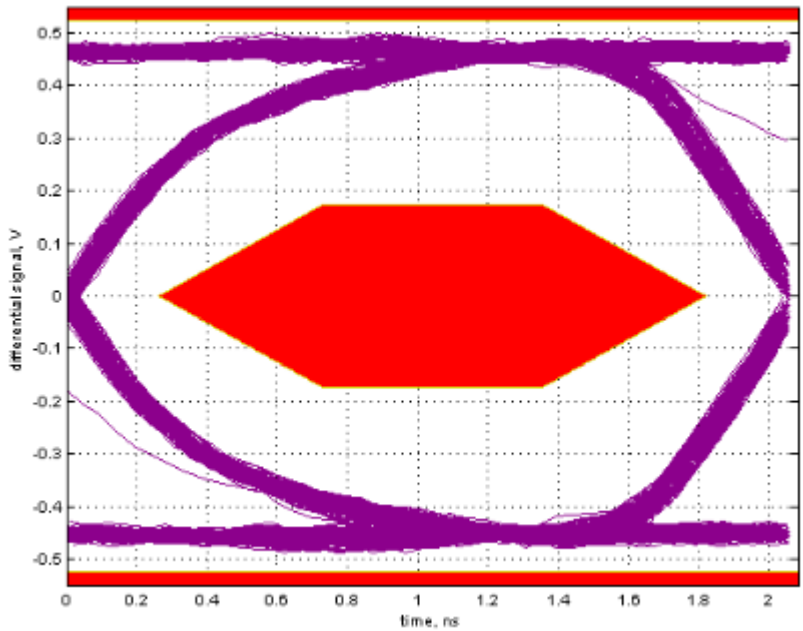


Figure 8 USB 2.0 High-speed Far End SQ Eye Diagram

From Figures 7 and 8, it can be seen that a standard USB 2.0 eye diagram appears as an "eye" with thin and clear traces, and the "eye" opens widely. When there is inter-symbol interference, the waveform is distorted, the symbols do not completely overlap, and the traces of the eye diagram become unclear, causing the "eye" to partially close. If noise is added, the lines of the eye diagram become blurred and the "eye" opens smaller. Therefore, the size of the "eye" opening indicates the degree of distortion and reflects the strength of inter-symbol interference.

1.5.2 USB 2.0 Eye Diagram Test Problem Analysis

1. Oscilloscope cannot detect the eye diagram test trigger signal

   First, check whether the test fixture is connected correctly and whether the USB test software on the oscilloscope is configured correctly. Observe from the oscilloscope whether the periodic test waveform as shown in Figure 3 is detected. If there is no periodic test waveform as in Figure 3, it may be that the test command was not executed successfully or the test command is incorrect.

2. The tested eye diagram is severely distorted

   Severe distortion of the tested eye diagram, such as amplitude distortion or signal collapse, is generally due to incorrect test operation methods.

As shown in Figure 9, the signal amplitude of the USB eye diagram is twice the standard value. If using the Agilent test suite, this is generally because the D+ and D- of the test fixture are not connected to the 50 ohm terminal SMA resistors.

As shown in Figure 10, there is an obvious collapse in the middle of the USB eye diagram signal. If using the Agilent test suite, this is generally because the switch on the test fixture was not set to the ON position.

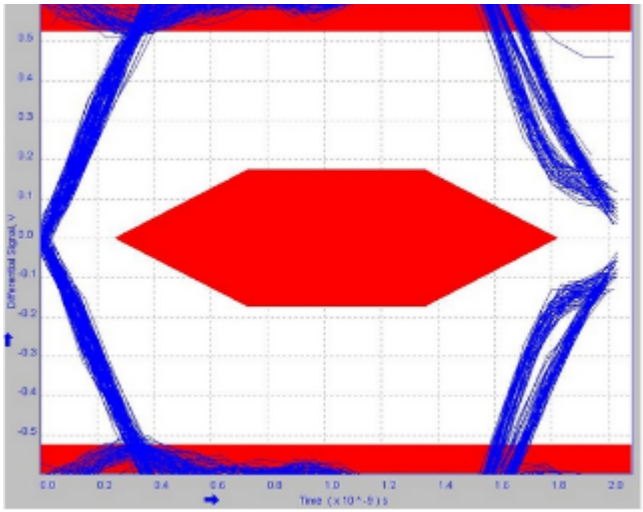


Figure 9 USB Eye Diagram Amplitude Distortion

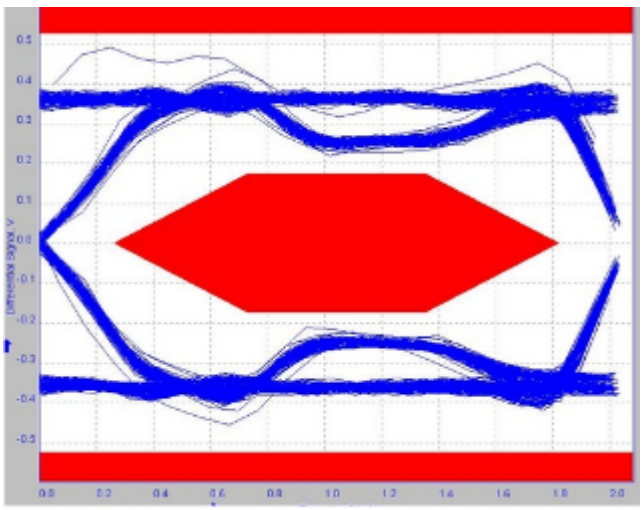


Figure 10 USB Eye Diagram Signal Collapse Distortion

3. USB eye diagram does not open

   As shown in Figure 11, the USB eye diagram does not open and will press against the USB eye diagram test template. Check whether ESD components or electronic switches with large internal capacitance are connected to the USB DP and DM lines. If so, remove these components and test again. You can also adjust the USB PHY drive strength and rise/fall edges through software. Please contact the engineer responsible for USB driver development for assistance in debugging.

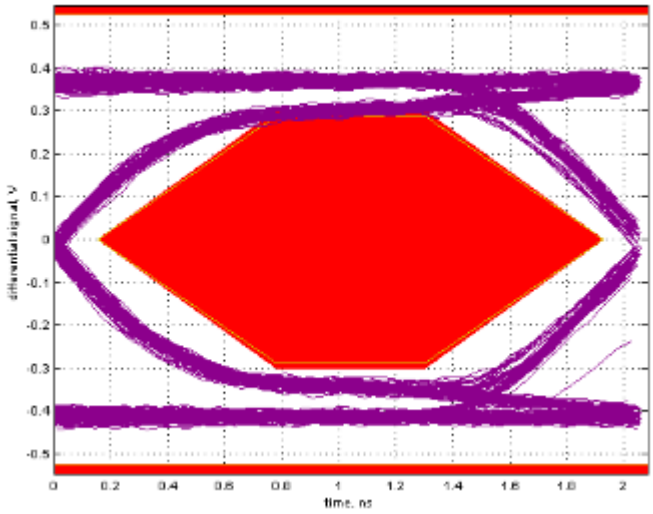


Figure 11 USB Eye Diagram Not Opening

4. USB eye diagram is blurred or even filled with bloodshot-like traces

   As shown in Figure 12, the outline of the USB eye diagram is blurred, indicating that the USB crosstalk is very severe, and there may also be impedance mismatch and noise interference issues. First, check whether ESD components or electronic switches with large internal capacitance are connected to the USB DP and DM lines. If so, remove these components and test again. Then, check whether the USB cable used for testing has impedance mismatch issues, or try a different USB cable for retesting. Finally, check the USB PCB routing, USB 24MHz clock source, and USB PHY power supply ripple.

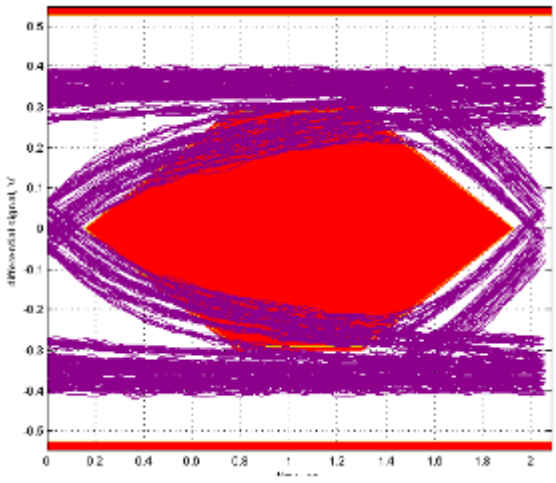


Figure 12 Blurred USB Eye Diagram

1.6 USB 2.0 Rx Compliance Test

1.6.1 USB 2.0 Receiver Sensitivity Test Principle

According to the test specification USB 2.0 Electrical Compliance Test Specification Version 1.07, USB 2.0 Rx test items include the following three:

EL_16 Receiver Sensitivity Test @ Squelch

EL_17 Receiver Sensitivity Test

EL_18 Receiver Sensitivity Test - Minimum SYNC Field

Test principle:

Set the Device Under Test (DUT) to enter SE0_NAK test mode. The signal generator simulates a USB HOST sending IN packet requests to the DUT. If the DUT Rx can properly receive the IN packet, the DUT will feed back a NAK packet. By changing the amplitude of the IN packet waveform output by the signal generator and detecting whether the DUT has NAK feedback, the DUT's receiver sensitivity can be measured.

EL_16 sweeps from high amplitude to low amplitude. If the signal generator is below 100mV and the DUT still has feedback, it means the receiver is too sensitive, fail.

EL_17 sweeps from low amplitude to high amplitude. If the signal generator is above 200mV and the DUT still does not have stable feedback, it means the receiver is too insensitive, fail.

Note:

- *The EL_17 specification was originally set to 150mV by the USB Spec, and was later changed to 200mV by ECN*
- *According to the test specification, the signal generator simulates the behavior of a USB HOST, so the DUT can only be a Device or HUB Upstream port. HOST does not support receiver sensitivity testing*
- *The NAK packet pattern is fixed: 32-bit sync header (31 bits "0" + 1 bit "1") + 8-bit (10100101) NAK PID encoding*

1.6.2 USB 2.0 Rx Test Commands

Table 5 USB 2.0 Test SE0_NAK Commands

Chip Name	USB 2.0 Test SE0_NAK Command
RK1808	io -4 0xfd00c704 0x8c000a06
RK1820	io -4 0x102500c704 0x8c000a06
RK2108	io -4 0x41300804 0x30
RK2116	io -4 0x40040804 0x30
RK2206	io -4 0x43040804 0x30
RK29XX RK30XX RK31XX	io -4 0x10180804 0x30
RK3228 RK3229	io -4 0x30040804 0x30
RK3288 RK3228H RK3328 RK3368	io -4 0xff580804 0x30
RK3308 RK3308B-S RK3308H-S	io -4 0xff400804 0x30
RK3326 RK3326S PX30 PX30S	io -4 0xff300804 0x30
RV1103 RV1106	io -4 0xffb0c704 0x8c000a06
RV1108	io -4 0x30180804 0x30
RV1126 RV1109	io -4 0xffd0c704 0x8c000a06
RV1126B	io -4 0x2150c704 0x8c000a06
RK3399	io -4 0xfe80c704 0x8c000a06
RK3566 RK3568	io -4 0xfcc0c704 0x8c000a06
RK625	io -4 0x40180804 0x30
RK3588 RK3588S	io -4 0xfc00c704 0x8c000a06

1.6.3 USB 2.0 Rx Test Method

For the standard USB 2.0 Rx test method, please refer to the test specification USB 2.0 Electrical Compliance Test Specification Version 1.07, Section 4.2.1 Upstream Port Receiver Sensitivity. This document uses the Tektronix test environment for illustration.

1.6.3.1 USB 2.0 Rx Test Notes

- During testing, you must first connect the DUT to the PC Host through the test fixture to prevent the USB Device PHY from entering suspend mode and affecting the test, and then input the test command;
- During testing, you need to input the Test SE0_NAK command via serial console to make the DUT USB controller enter Test_SE0_NAK mode;
- During the EL_18 Receiver Sensitivity Test - Minimum SYNC Field test, when the Tektronix signal generator switches from 32-bit SYNC to 12-bit SYNC, you need to swap the channel polarity of the signal generator.

1.6.3.2 USB 2.0 Rx Test Steps

1. As shown in Figure 13 below, connect the DUT to the Tektronix signal generator and oscilloscope through the Device Sensitivity fixture.

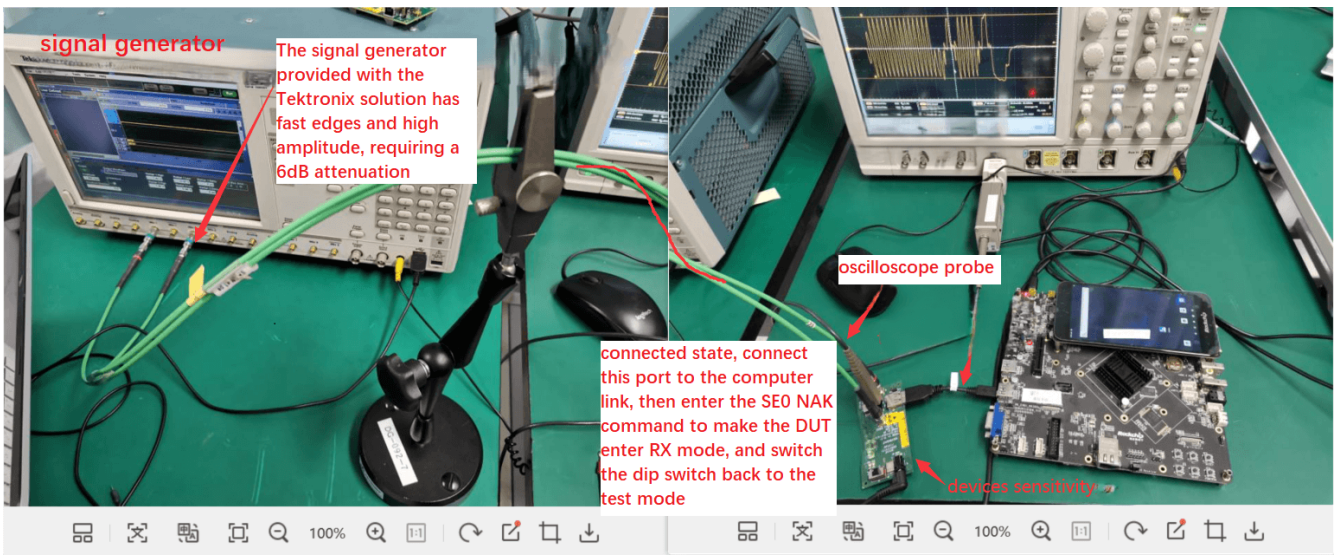


Figure 13 Tektronix USB 2.0 Rx Test Connection

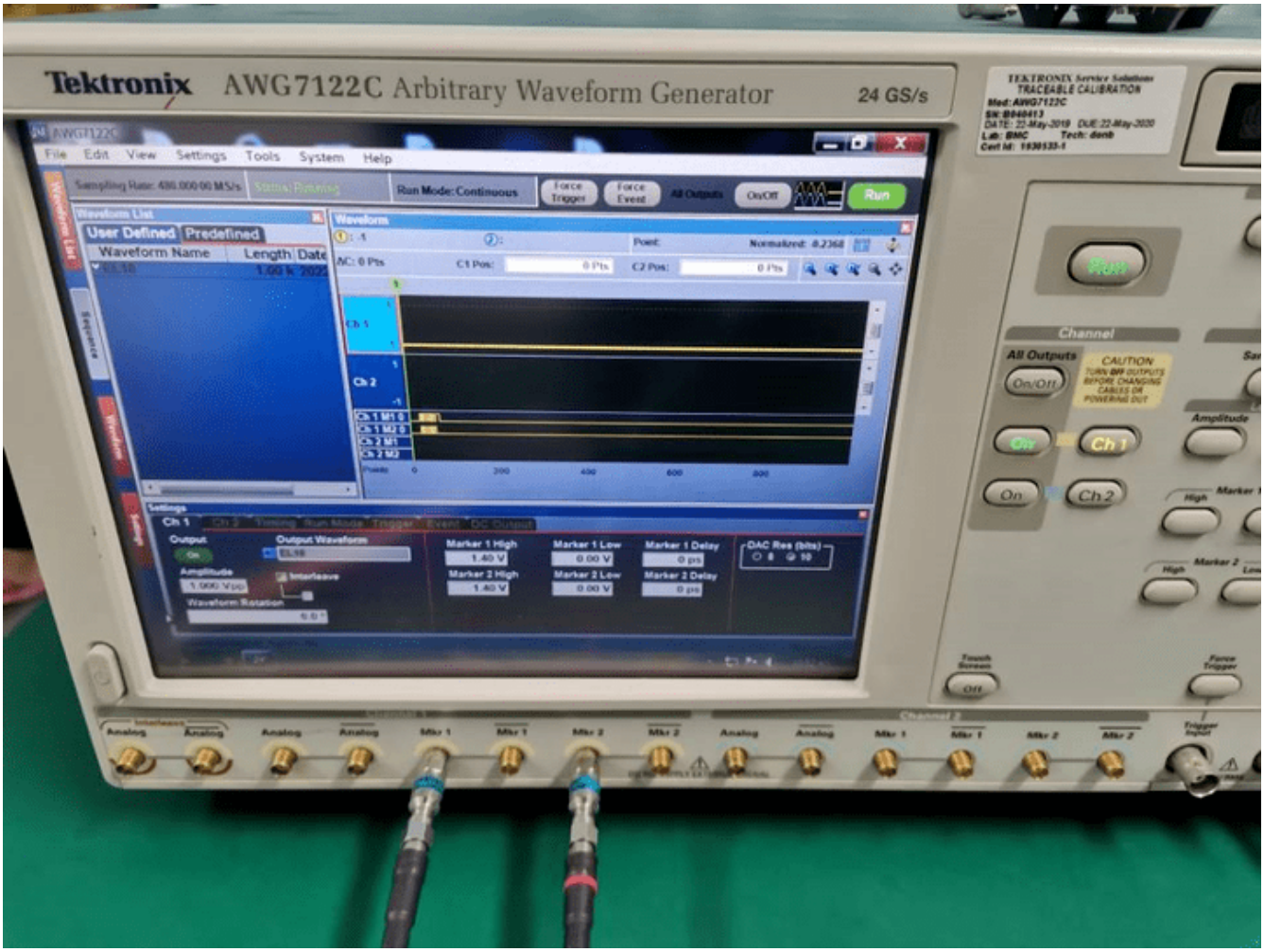


Figure 14 Tektronix Signal Generator

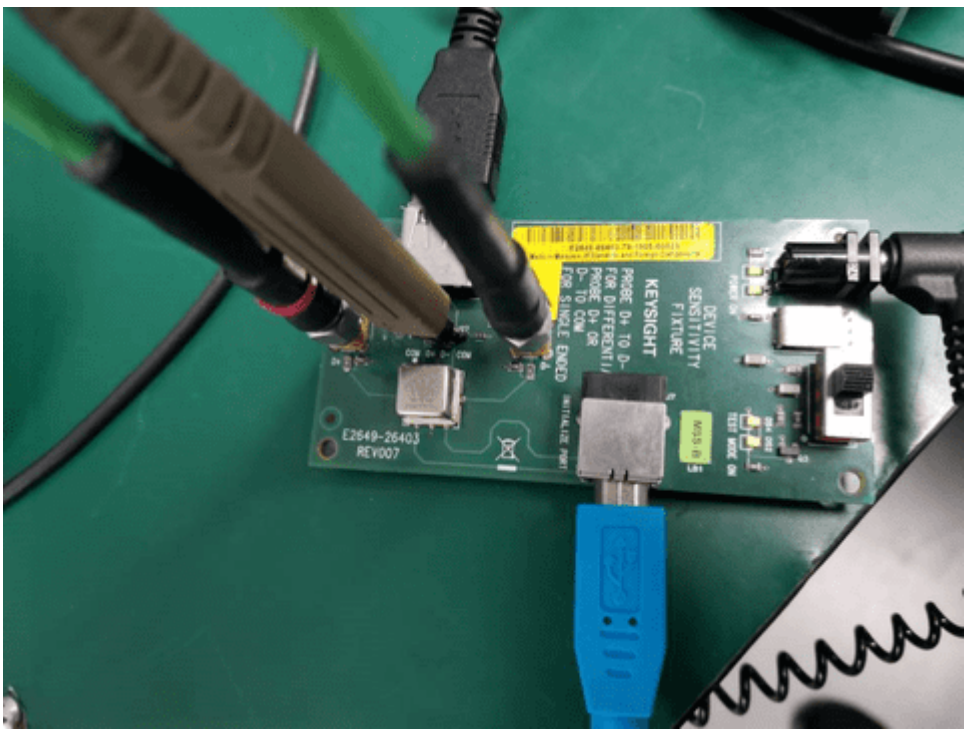


Figure 15 USB 2.0 Rx Device Sensitivity Fixture

2. Switch the DIP switch of the USB 2.0 Rx Device Sensitivity fixture to ON, then connect the test fixture to the PC Host interface
3. Look up Table 5 USB 2.0 Test SE0_NAK commands, and input the command via serial console to make the DUT enter SE0_NAK mode
4. Switch the DIP switch of the USB 2.0 Rx Device Sensitivity fixture to OFF
5. Set the Tektronix signal generator to output the test pattern, and complete the EL_16/EL_17/EL_18 test items in sequence according to the test procedure

1.6.3.3 USB 2.0 Rx Test Result Analysis

As shown in Figure 16 below, the Tektronix oscilloscope samples and tests the IN packet waveform sent by the signal generator and the NAK packet waveform fed back by the DUT in real time, and finally generates the test report shown in Figure 17.

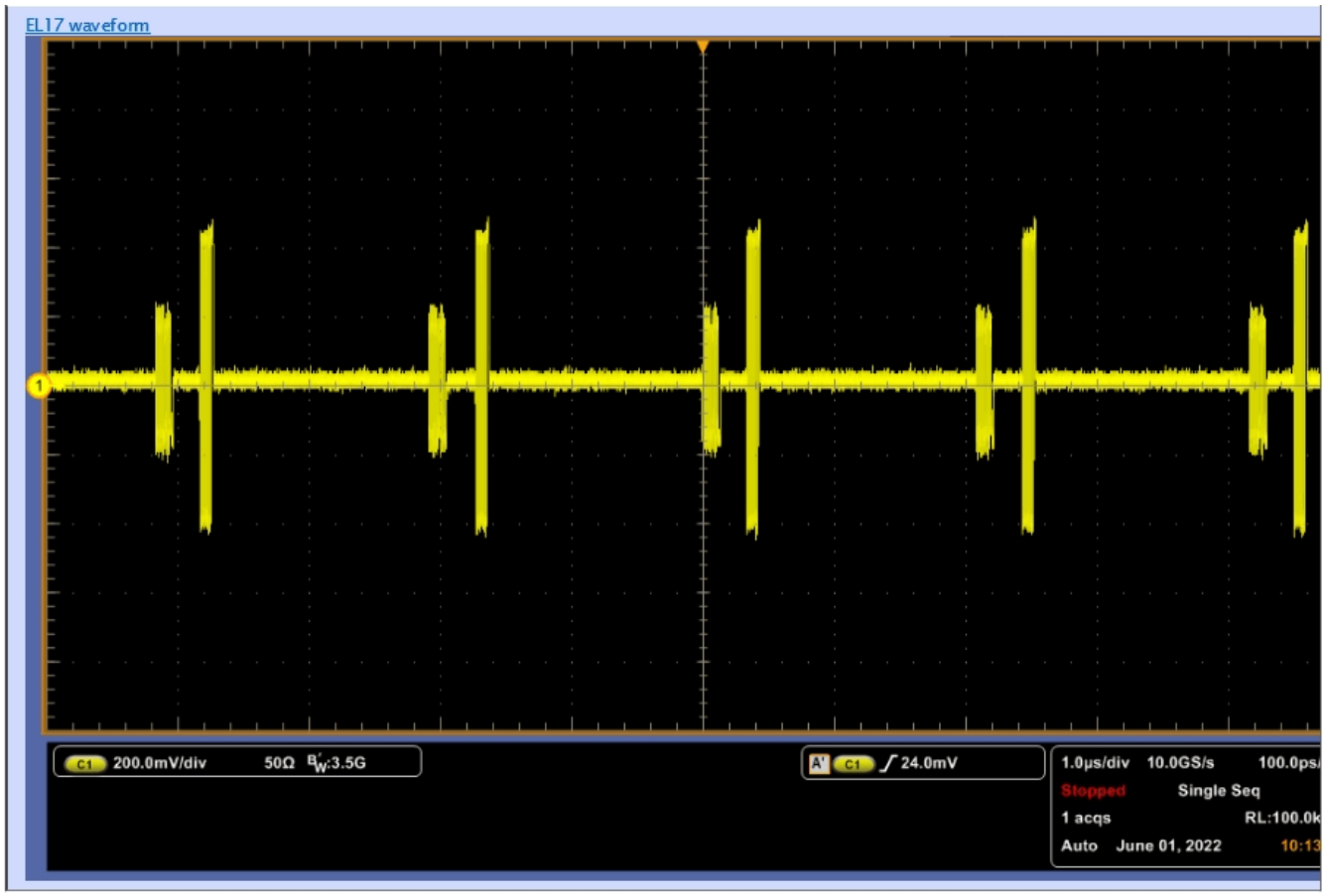
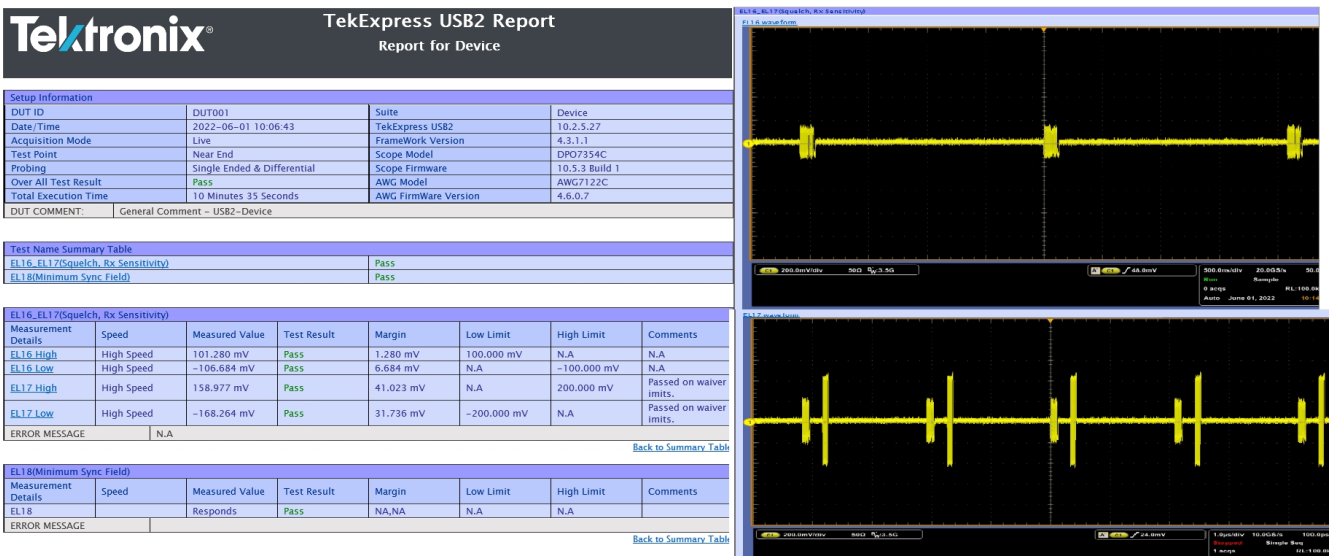


Figure 16 USB 2.0 Rx Test Signal



TekExpress USB2 Report

Report for Device

Setup Information			
DUT ID	DUT001	Suite	Device
Date/Time	2022-06-01 10:06:43	TekExpress USB2	10.2.5.27
Acquisition Mode	Live	FrameWork Version	4.3.1.1
Test Point	Near End	Scope Model	DPO7354C
Probing	Single Ended & Differential	Scope Firmware	10.5.3 Build 1
Over All Test Result	Pass	AWG Model	AWG7122C
Total Execution Time	10 Minutes 35 Seconds	AWG FirmWare Version	4.6.0.7
DUT COMMENT:	General Comment – USB2-Device		

Test Name Summary Table	
EL16_EL17(Squelch, Rx Sensitivity)	Pass
EL18(Minimum Sync Field)	Pass

EL16_EL17(Squelch, Rx Sensitivity)

Measurement Details	Speed	Measured Value	Test Result	Margin	Low Limit	High Limit	Comments
EL16 High	High Speed	101.280 mV	Pass	1.280 mV	100.000 mV	N.A	N.A
EL16 Low	High Speed	-106.684 mV	Pass	6.684 mV	N.A	-100.000 mV	N.A
EL17 High	High Speed	158.977 mV	Pass	41.023 mV	N.A	200.000 mV	Passed on waiver imits.
EL17 Low	High Speed	-168.264 mV	Pass	31.736 mV	-200.000 mV	N.A	Passed on waiver imits.
ERROR MESSAGE	N.A						

Back to Summary Table

EL18(Minimum Sync Field)

Measurement Details	Speed	Measured Value	Test Result	Margin	Low Limit	High Limit	Comments
EL18		Responds	Pass	NA,NA	N.A	N.A	
ERROR MESSAGE							

Back to Summary Table

Figure 17 USB 2.0 Rx Test Report

When the test results EL_16/EL_17 fail, you can try to resolve it by adjusting the Squelch parameter of the USB 2.0 PHY. For the specific method, please refer to the section Rockchip USB SQ Tool.

2. USB 2.0 HUB SQ Test

The SQ test of USB 2.0 HUB includes upstream ports and downstream ports. However, in practical applications, we generally only need to test the downstream ports provided for user use. Therefore, this document only provides the SQ test method for testing USB 2.0 HUB downstream ports.

Common USB 2.0 HUB chip models mainly include: FE1.1, FE1.1S, GL850, GL852, USX2064, HX2VL, etc. Common HSIC HUB models mainly include: USB4604, GL850H. For USB 2.0 HUB, there are two test methods on the Rockchip platform, see Method 1 - Command Test and Method 2 - Script Test below. For **HSIC HUB**, only Method 2 - Script Test can be used.

Method 1 - Command Test

The test commands used in this method are the same as those in "USB 2.0 Host Tx Test Commands". The test steps are as follows:

- Identify the USB HOST controller connected to the HUB, then look up Table 2, Table 3, and Table 4 to find the test command corresponding to the HOST controller
- Refer to "USB 2.0 Tx Test Steps" to complete the SQ test of all downstream ports of the HUB

Note: For different HUB downstream ports, the test commands and test methods are the same.

Method 2 - Script Test

Compared with Method 1 - Command Test, Method 2 - Script Test is more complex and requires compiling and running a script, but it is more universal and can test all types of HUBs, including USB 2.0 HUB and HSIC HUB.

1. Download and compile the test script

   Script source code download link: https://redmine.rockchip.com.cn/documents/113

File: USB_HUB_Compliance_Test_Script

2. Execute the test script

   Copy the compiled executable file linux-eye to the data directory of the system, and execute the command

   `chmod 777 linux-eye`

   Execute the test script linux-eye, then follow the script prompts to input test commands, as shown below:

```
[root@hari LinuxEye]# ./linux-eye
LinuxEye - select one of the following hub for testing.
[ 0] 4-port Full-Speed hub at tier 2 of Bus 3
(VID: 0451, PID: 1446, Address: 3)
[ 1] 4-port High-Speed hub at tier 2 of Bus 1
(VID: 1A40, PID: 0101, Address: 15)
[ 2] 4-port High-Speed hub at tier 2 of Bus 1
(VID: 1A40, PID: 0101, Address: 10)
[ 3] 7-port High-Speed hub at tier 3 of Bus 1
(VID: 1A40, PID: 0201, Address: 50)
Please enter [0 ~ 3] to select a hub or 'q' to quit: 2 (Indicates there are 4 HUBs in
total, testing HUB[2])
[ 1] is connected to Low-Speed device
[ 2] is open
[ 3] is connected to High-Speed device
[ 4] is connected to Low-Speed device
Please enter [1 ~ 4] to select a port or 'q' to quit: 2 (Indicates testing port 2 of the
HUB)
LinuxEye - Start testing port 2 of device 10 on bus 1
Type 'q' to stop the test: q (Quit the test script)
[root@hari LinuxEye]#
```

3. USB 3.0 Compliance Test

USB 3.0 is a dual-bus architecture that adds the Super Speed bus portion on top of USB 2.0. The Super Speed bus has a signal rate of 5Gbps, uses ANSI 8b/10b encoding, operates in full-duplex mode, and supports a maximum cable length of 3 meters. The following Figure 18 shows a typical USB 3.0 bus architecture.

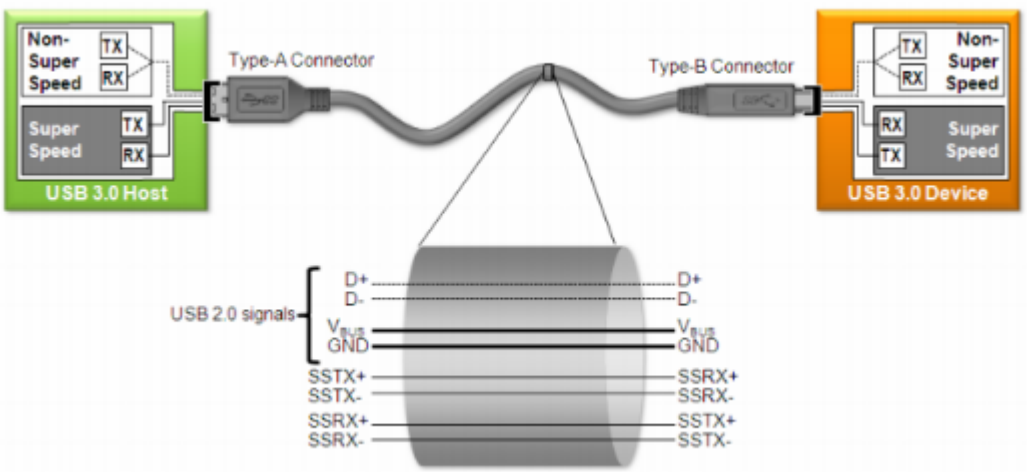


Figure 18 USB 3.0 Bus Architecture

3.1 New USB 3.0 Test Specifications

- Compliance calibration and testing are performed at the end of the compliance channel

  The compliance channel is used to characterize the worst-case interconnect channel condition during TX and RX testing

  Host: **3 meter cable + 5 inch trace**

  Device: **3 meter cable + 11 inch trace**
- TX testing allows channel embedding, selecting golden S-parameters for embedding testing
- Need to calculate DJ, RJ, and TJ based on 1e-12 bit error rate

  Added 10MHz, 20MHz, and 33MHz compliance Pj test frequency points
- Post-processing requires the use of a CTLE equalizer, and the eye diagram and its parameters are observed and analyzed after the equalizer. Since the eye diagram of a 5Gbps signal may not open after transmission through long cables and PCB, the USB 3.0 chip receiver internally provides CTLE (Continuous Time Linear Equalization) functionality to compensate for high-frequency loss. Therefore, the oscilloscope's test software must also support CTLE to obtain accurate results.
- Device receiver eye diagram amplitude calibration standard is 145mVp-p
- Host receiver eye diagram amplitude standard is 180mVp-p

  The electrical performance testing of USB 3.0 is divided into **transmit signal test (Tx)**, **receive tolerance test (Rx Tolerance Compliance Test)**, and cable/connector testing.

3.2 USB 3.0 Tx Compliance Test

3.2.1 USB 3.0 Tx Test Requirements

During transmitter testing, the test object is required to emit specific test patterns, and the real-time oscilloscope performs eye diagram analysis on the pattern, measuring the signal amplitude, jitter, average data rate, and rise/fall times. USB 3.0 signal testing for the SuperSpeed portion is significantly different from the previous USB 2.0 test method.

First, since the USB 3.0 SuperSpeed signal rate reaches 5Gbps and the signal amplitude is smaller, the test requires an oscilloscope with **bandwidth above 12GHz**, and the oscilloscope must have lower noise floor to ensure accurate measurement.

Second, USB 3.0 transmitter testing does not connect the DUT directly with a fixture. The defined test point is the end of the "**Compliance Channel**". The compliance channel simulates the effect of PCB traces and cables on the signal. For HOST testing, it simulates the effect of a 3m long cable + 5 inch PCB trace; for Device testing, it simulates the effect of a 3m long cable + 11 inch PCB trace. The USB 3.0 test specification provides the compliance channel model in the form of S-parameter files. During actual testing, the test fixture is directly connected to the DUT, and the channel effect is added through S-parameter embedding on the oscilloscope. As shown in the Tx test model in Figure 19, TP1 is the oscilloscope test point.

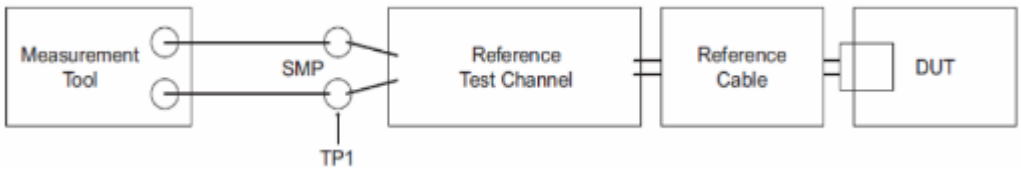


Figure 19 USB 3.0 Tx Test Model

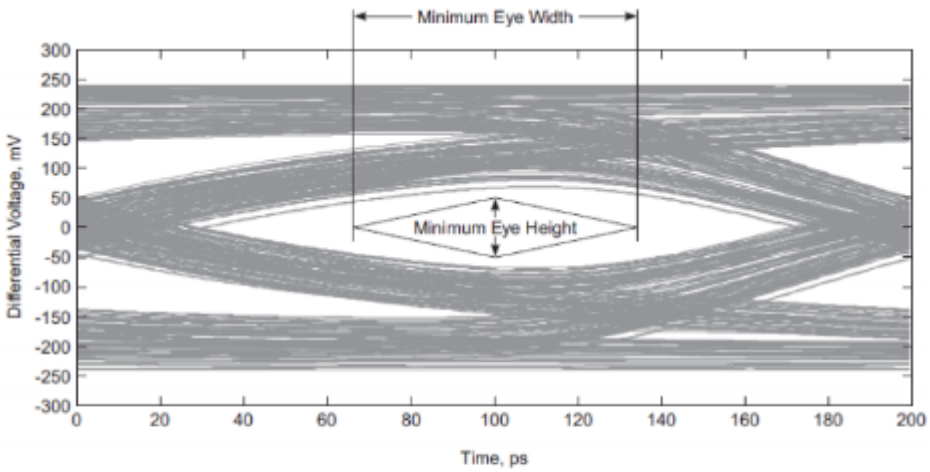


Table 6-12. Normative Transmitter Eye Mask at Test Point TP1

Signal Characteristic	Minimal	Nominal	Maximum	Units	Note
Eye Height	100		1200	mV	2, 4
Dj			0.43	UI	1,2,3
Rj			0.23	UI	1,2,3, 5
Tj			0.66	UI	1,2,3

Figure 20 USB 3.0 Tx Test Eye Diagram Requirements

Table 6-10. Transmitter Normative Electrical Parameters

Symbol	Parameter	5.0 GT/s	Units	Comments
UI	Unit Interval	199.94 (min) 200.06 (max)	ps	The specified UI is equivalent to a tolerance of ±300 ppm for each device. Period does not account for SSC induced variations.
$V_{TX-DIFF-PP}$	Differential p-p Tx voltage swing	0.8 (min) 1.2 (max)	V	Nominal is 1 V p-p
$V_{TX-DIFF-PP-LOW}$	Low-Power Differential p-p Tx voltage swing	0.4 (min) 1.2 (max)	V	Refer to Section 6.7.2. There is no de-emphasis requirement in this mode. De-emphasis is implementation specific for this mode.
$V_{TX-DE-RATIO}$	Tx de-emphasis	3.0 (min) 4.0 (max)	dB	Nominal is 3.5 dB

Figure 21 USB 3.0 Tx Test Electrical Parameter Requirements

3.2.2 USB 3.0 Tx Test Items

- LFPS (Near End)
- SSC (Near End)
- Tx (Near End/Far End): Eye diagram; Tj, Rj, Dj; Amplitude;

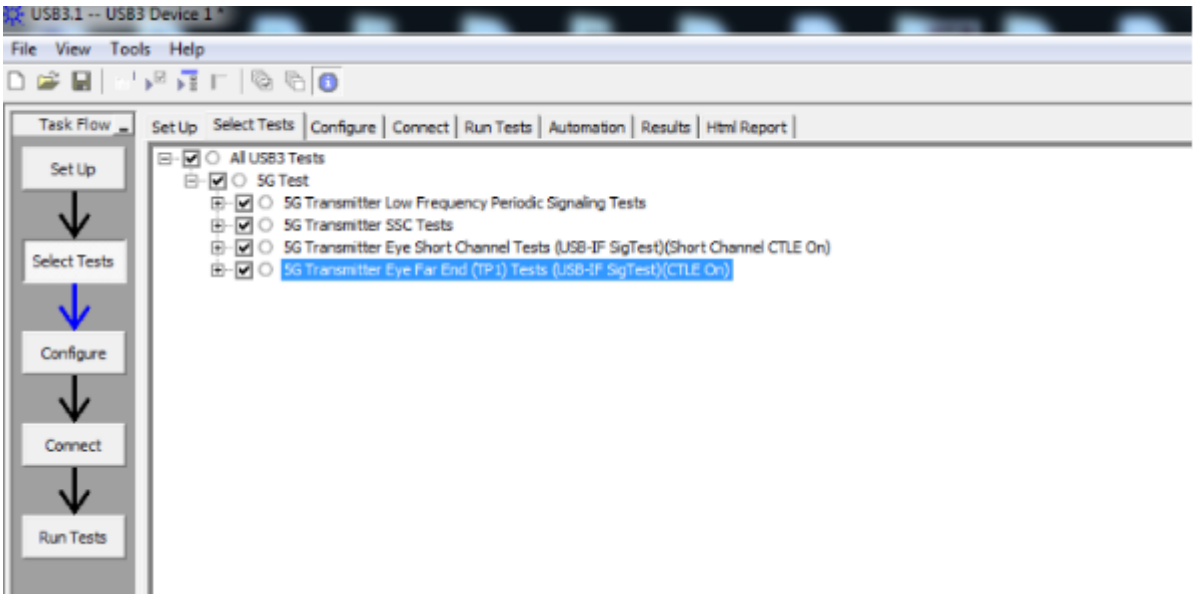


Figure 22 Agilent USB 3.0 Tx Test Options

3.2.3 USB 3.0 Tx Test Mode

According to the USB 3.0 spec, the USB 3.0 controller must first enter Compliance Mode before USB 3.0 Tx signal integrity testing can begin. As shown in Figure 23, after the **first LFPS timeout** in the Polling phase, the controller will exit from Polling.LFPS to Compliance Mode.

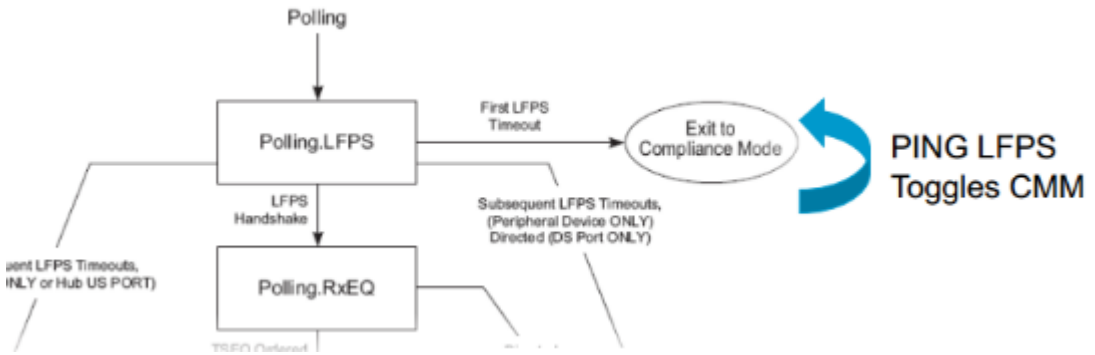


Figure 23 USB 3.0 Flow to Enter Compliance Mode

3.2.4 USB 3.0 Tx Test Environment

Agilent USB 3.0 Tx Test Suite

For USB 3.0 Tx signal testing, Agilent recommends using the 90000 series oscilloscope (providing up to 13GHz bandwidth) with the automated compliance test software U7243A and test fixture U7242A to complete the USB 3.0 specification transmitter testing and verification.

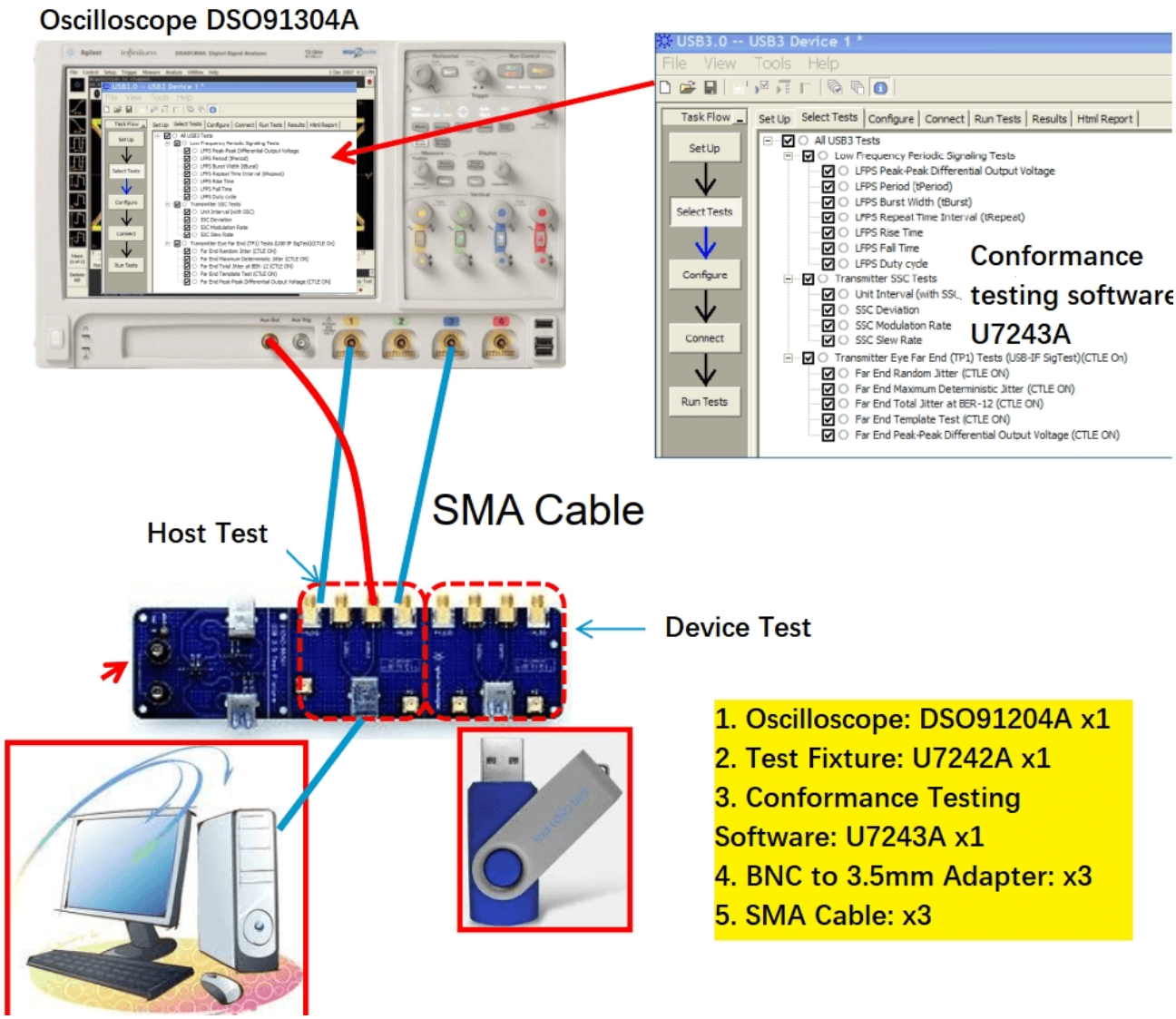


Figure 24 Agilent USB 3.0 Tx Test Environment

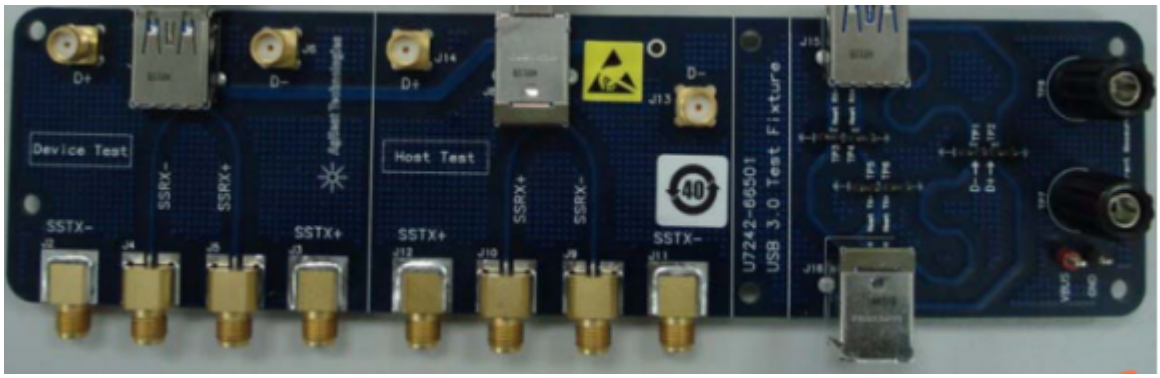


Figure 25 Agilent USB 3.0 Tx Test Fixture U7242A

In addition, Agilent also provides the USB 3.1 Gen1 Type-C test fixture N7015A, as shown in Figure 26 below, for testing the signal integrity of USB 3.0/3.1 Type-C interfaces. The test software is the same as for the U7242A fixture. For the specific usage of the Type-C test fixture, please refer to the document "Keysight N7015A-16A Type-C Test Kit".

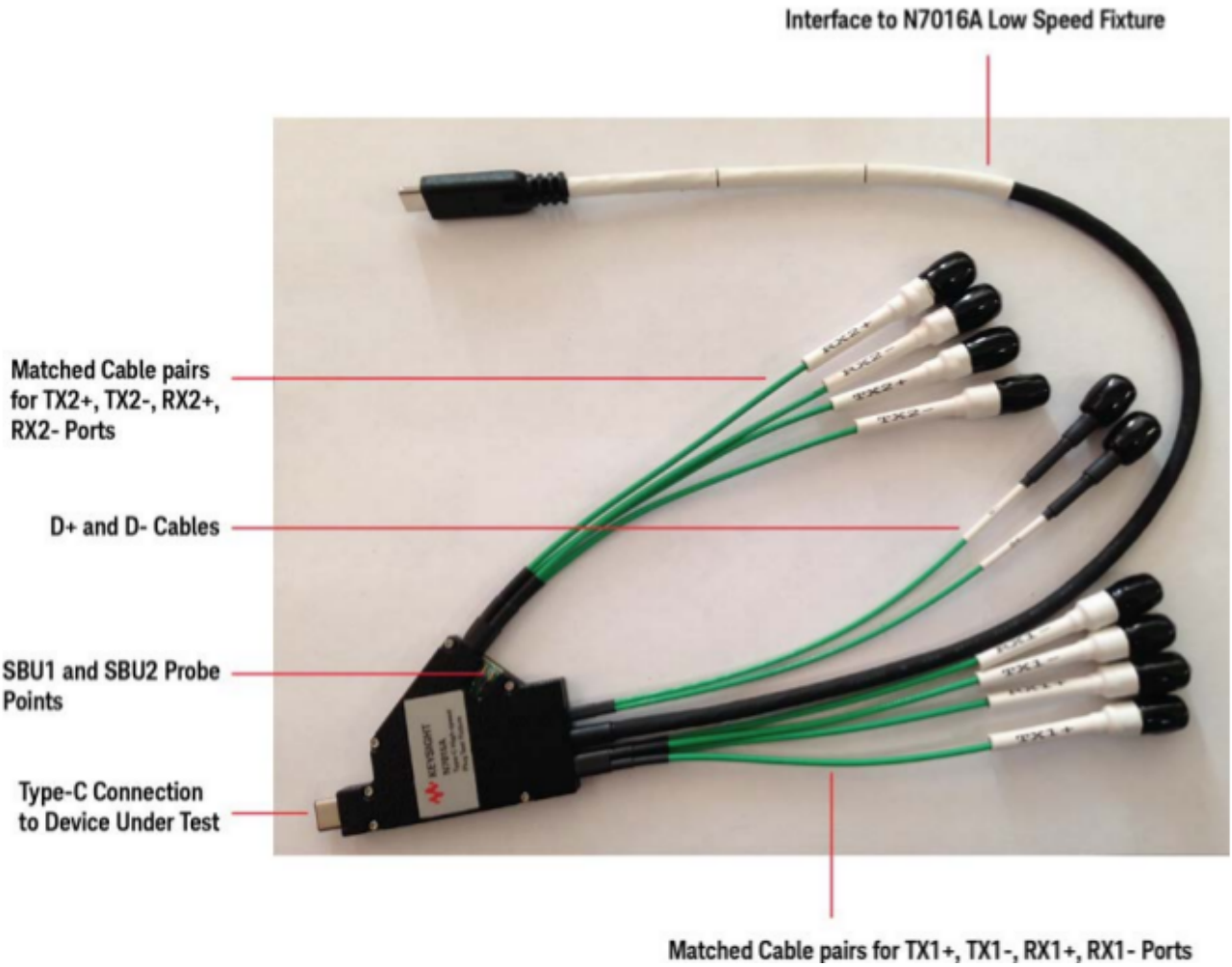


Figure 26 Agilent USB 3.0 Type-C Test Fixture N7015A

Tektronix USB 3.0 Tx Test Suite

The Tektronix Tx test connection diagram is shown in Figure 27. The Tektronix USB 3.0 transmitter measurement (option USB-Tx) is applicable to DPO/MSO70000 series oscilloscopes and provides an automated USB 3.0 transmitter solution.

For specific test solutions, please refer to:

https://www.tek.com.cn/datasheet/usb3-transmitter-and-receiver-solutions-datasheet

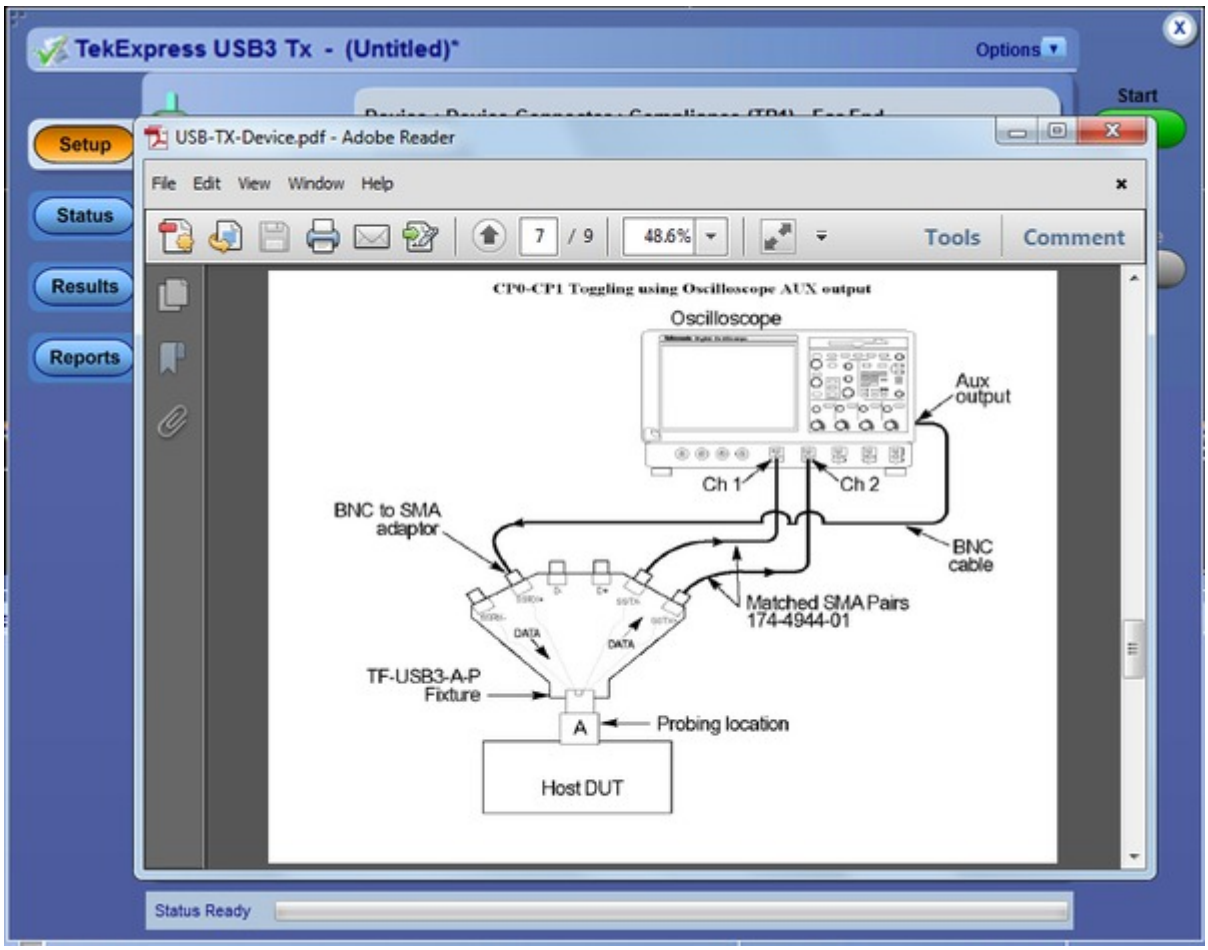


Figure 27 Tektronix USB 3.0 Tx Test Diagram

3.2.5 USB 3.0 Device Tx Test Commands

The following chips require test commands to be input when testing USB 3.0 Device Tx, in order to trigger the CP1 test pattern. Moreover, **you must strictly follow the test step requirements and input the Device test command when the oscilloscope prompts to test the CP1 test pattern**. Otherwise, it may cause abnormal testing of the CP0 test pattern or CP1 test pattern.

Table 6 USB 3.0 Device Tx Test Commands

Chip Name	DWC3_0 Device USB 3.0	DWC3_1 Device USB 3.0
RK1808	io -4 0xff384008 0xc	N.A
RK3228H RK3328	io -4 0xff478408 0xc	N.A

Note: Under normal circumstances, USB 3.0 Device does not need to input test commands and can automatically enter Tx test mode. The RK1808/RK3228H/RK3328 test commands listed in Table 6 are intended to manually trigger the CP1 test pattern. Other chips, such as RK3399/RK3568/RK3588, do not need to execute Device Tx test commands.

3.2.6 USB 3.0 Device Tx Test Method

This document mainly describes the USB 3.0 Device Tx test method using the Agilent 90000 series oscilloscope (model: DSO91204A and test fixture U7242A). If you are using Tektronix or LeCroy oscilloscopes, please search for the test guides published officially by Tektronix and LeCroy.

3.2.6.1 USB 3.0 Device Tx Test Notes

- When testing USB 3.0 Device Tx, please first check Table 6 to confirm whether a test command needs to be input.

  If no test command is needed, simply follow the test steps prompted by the oscilloscope test software to connect the Device USB port to be tested to the test fixture, and the USB 3.0 controller will automatically enter Compliance mode.

  If a test command is needed, first follow the test steps prompted by the oscilloscope test software to connect the Device USB port to be tested to the test fixture, and the USB 3.0 controller will automatically enter Compliance mode. Then, when the oscilloscope prompts to test the CP1 test pattern, input the Device test command.

- When testing USB 3.0 Device Tx, **VBUS 5V must not be self-powered, otherwise the USB 3.0 controller will not be able to enter Compliance mode**.

  The VBUS power supply needs to be provided by the test fixture U7242A. You can connect the USB power port of the test fixture to the USB port of the oscilloscope or PC via a USB cable to provide VBUS 5V power.

3.2.6.2 USB 3.0 Device Tx Test Steps

1. Automated Test Software Setup

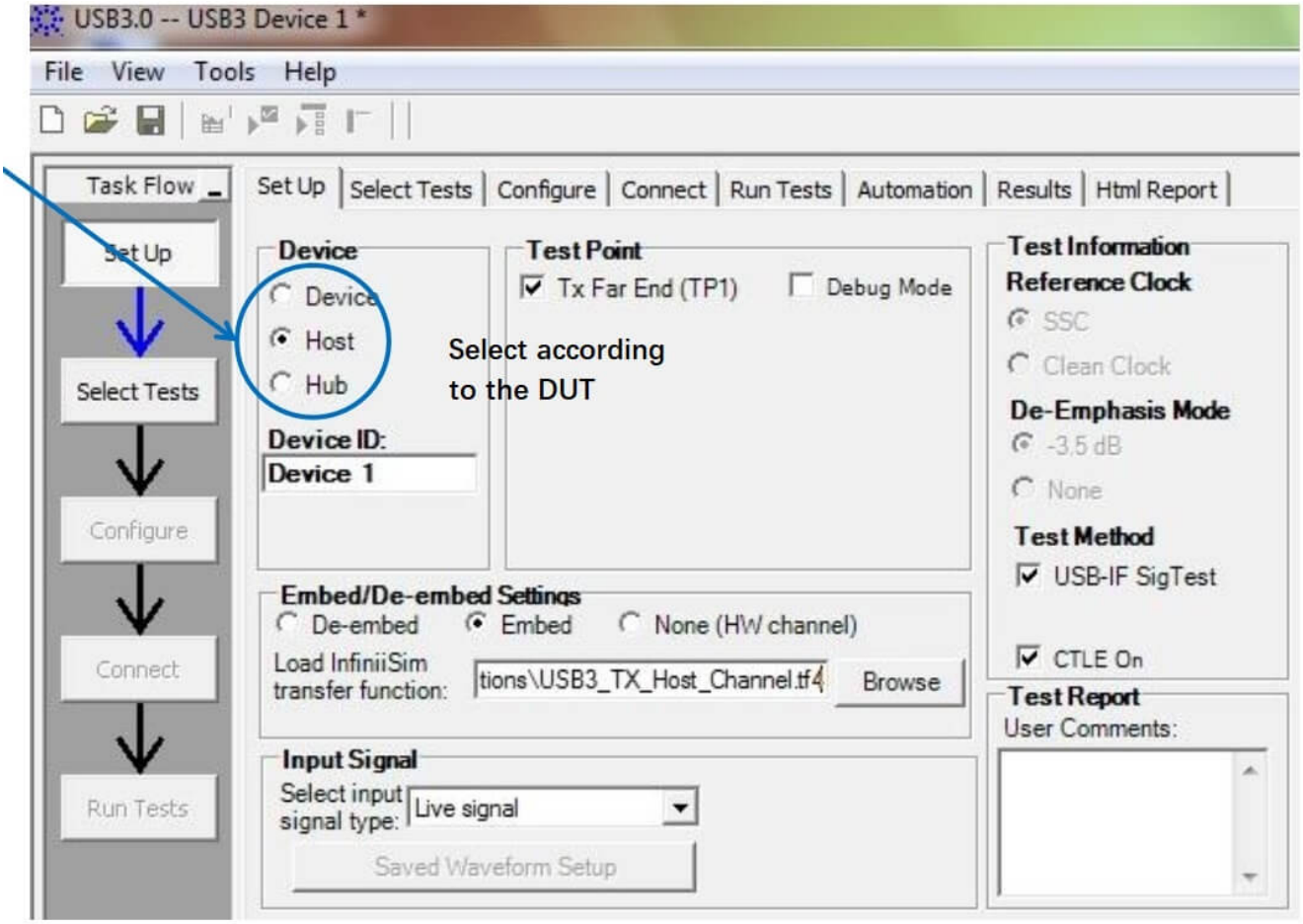
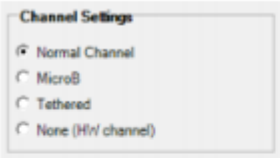


Figure 28 USB 3.0 Device Tx Test Software Setup Interface

Note:

The test software version used is ***V2.01****. If using a newer test version (such as V3.00.0001), the software setup method will be different.*

The Channel Setting method is shown in the figure below:



- **Normal Channel** – This option turns on the InfiniiSim function on the channel under test that uses a standard super speed connector.
- **MicroB** – This option turns on the InfiniiSim function on the channel under test that uses a MicroB super speed connector.
- **Tethered** – This option turns on the InfiniiSim function on the channel under test that has an attached cable.
- **None** – No embed or de-embed takes place and the original differential source is used to perform the analysis.

Figure 29 Channel Setting Method in USB 3.0 Device Tx Test Software

Channel Settings defaults to Normal Channel, which embeds S-parameters to simulate the effect of a 3m long USB cable + 5'' PCB trace. Therefore, it is required to connect the Device Under Test (DUT) directly with the test fixture during testing. If the USB cable used for testing is too long (greater than 10 cm), it may cause Far End test items to fail. It is recommended to select None for Channel Settings, or use a short cable less than 10cm for testing.

2. Select Test Items

Check All USB3 Tests to select all USB 3.0 Tx compliance test items.

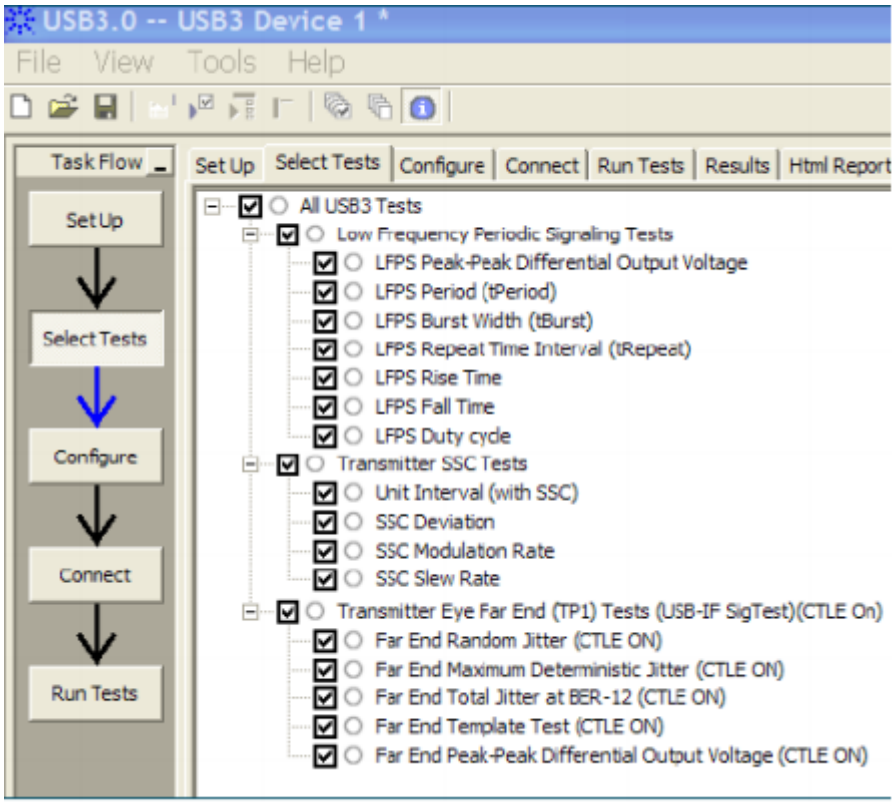


Figure 30 USB 3.0 Device Tx Test Item Settings

3. Configure Test Conditions

Set Automate Test Pattern Change to Auto, and use the default configuration for the rest.

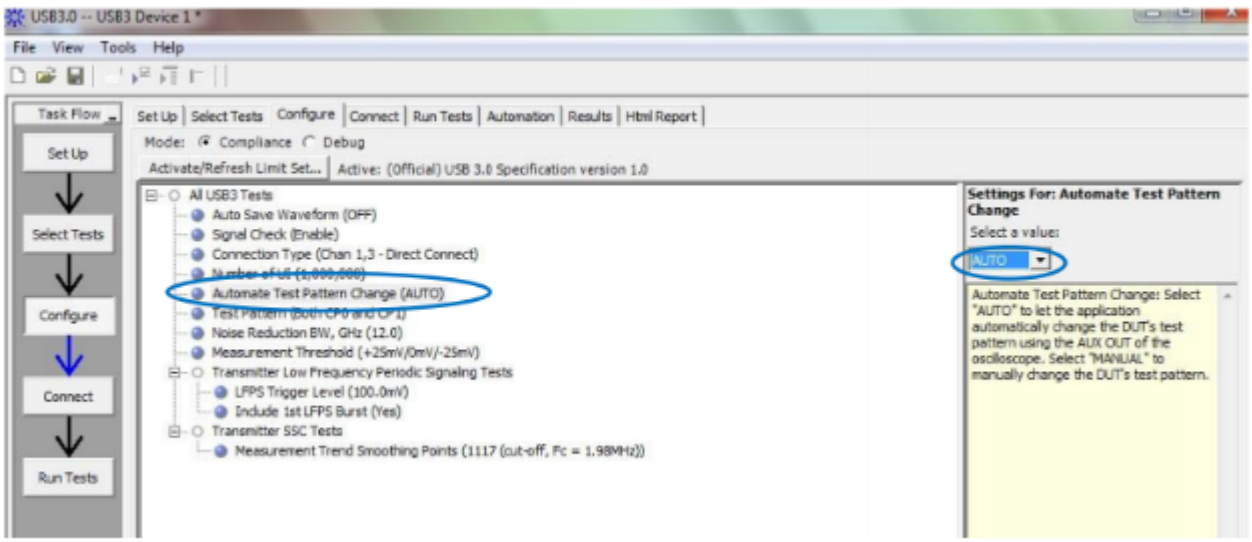


Figure 31 USB 3.0 Device Tx Test Condition Settings

4. Connect Oscilloscope, Fixture, and USB Device Under Test

Connect according to the oscilloscope instructions, as shown in the figure below. VBUS 5V power supply must also be connected.

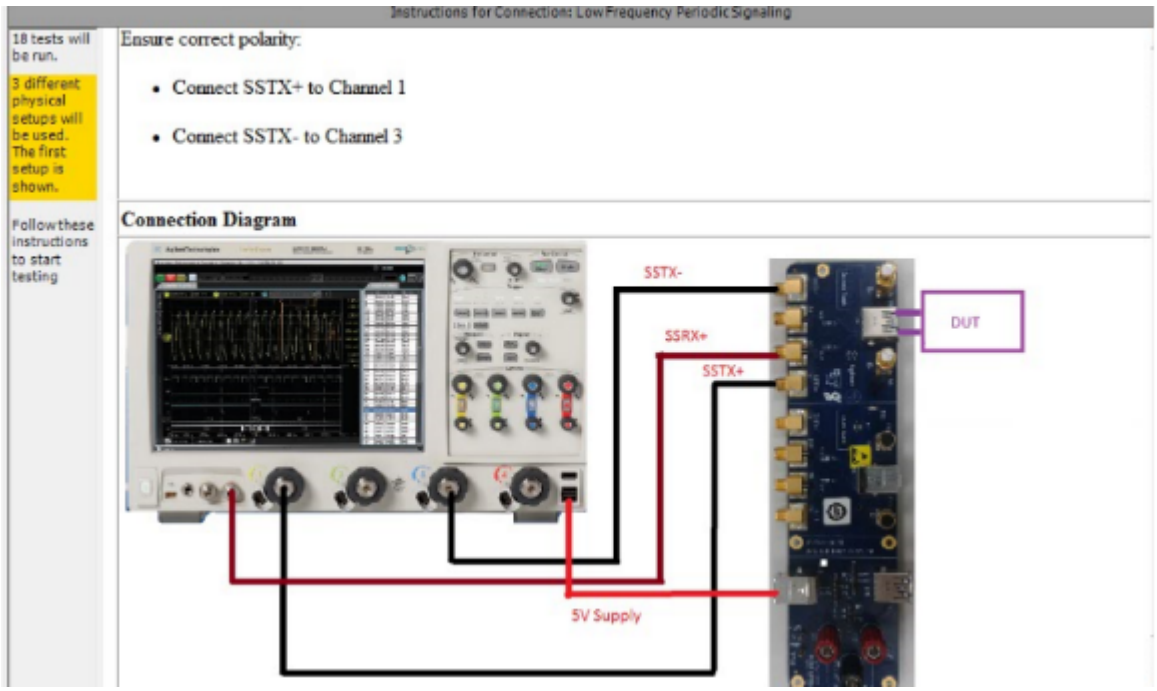


Figure 32 USB 3.0 Device Tx Test Connection Diagram

5. Start Tx Test

5.1. During the test, the automated software prompts the operation method for testing LFPS

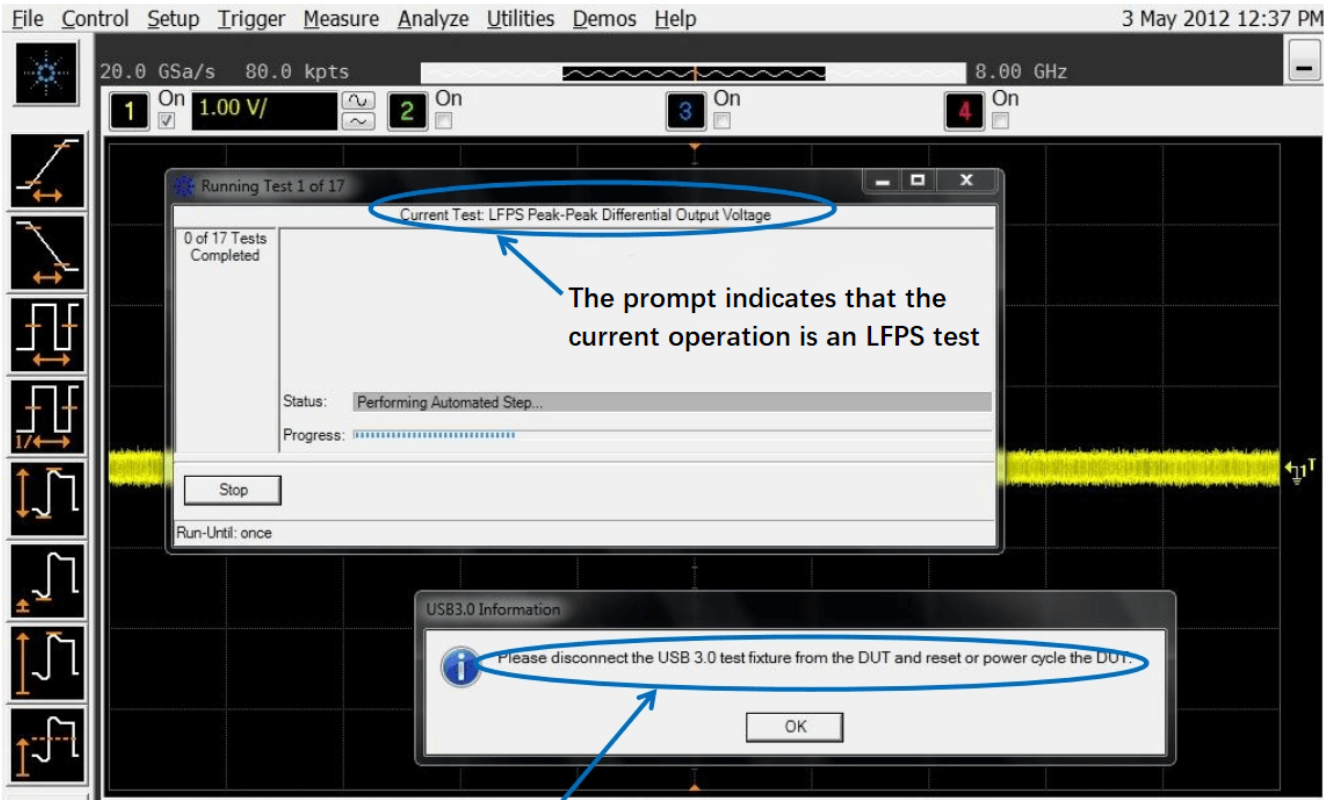


Figure 33 LFPS Test Interface

Note: Before performing the LFPS test, first disconnect the USB 3.0 fixture and the device under test, then click "OK", and reconnect to the fixture.

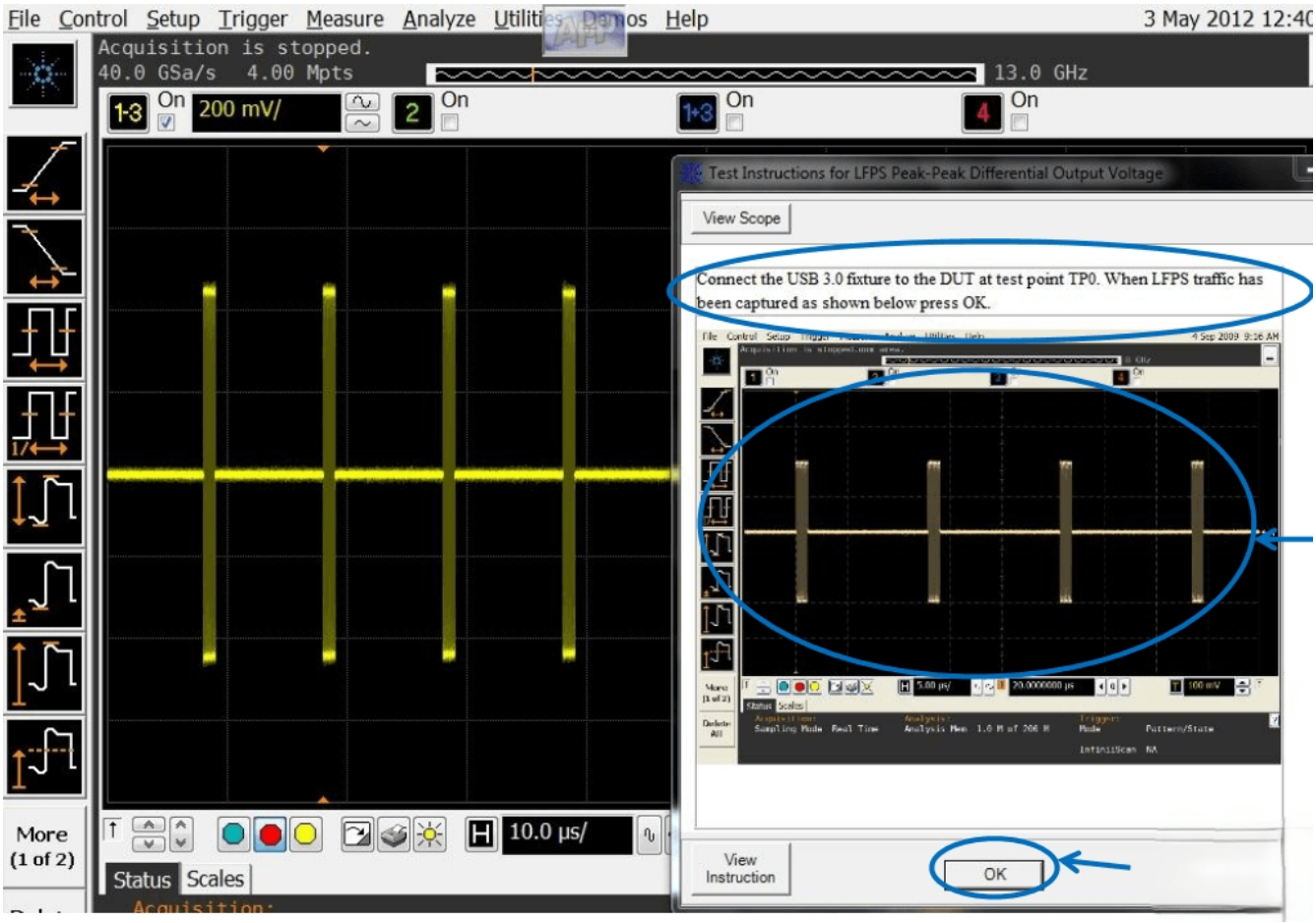


Figure 34 USB 3.0 Device Tx LFPS Reference Waveform

5.2. After the LFPS test is completed, the SSC test begins. The automated software prompts to change the connection of the oscilloscope, fixture, and device under test.

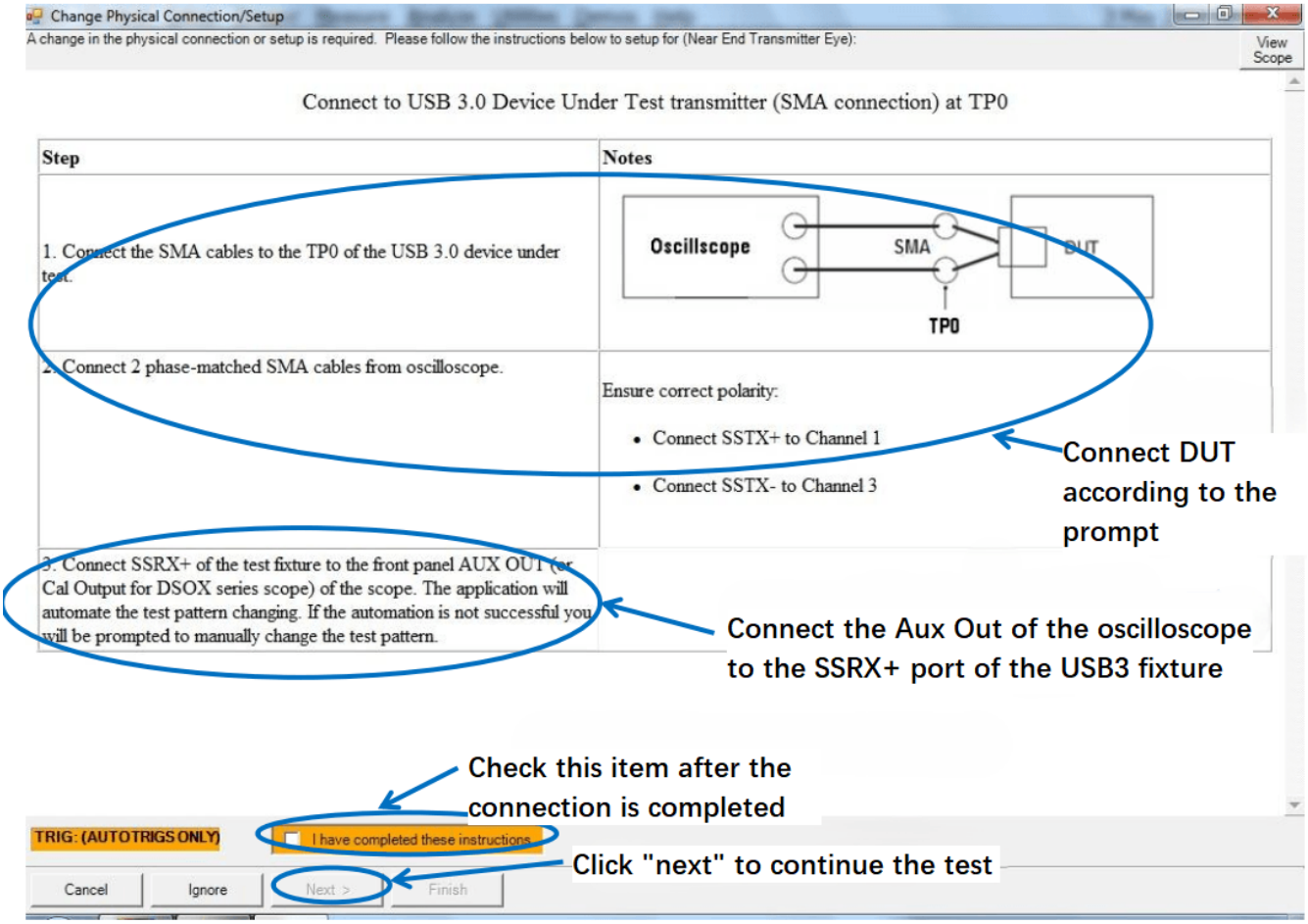


Figure 35 USB 3.0 Device Tx SSC Test

5.3. After the SSC test is completed, the eye diagram/jitter test begins. The automated software prompts to change the connection of the oscilloscope, fixture, and device under test.

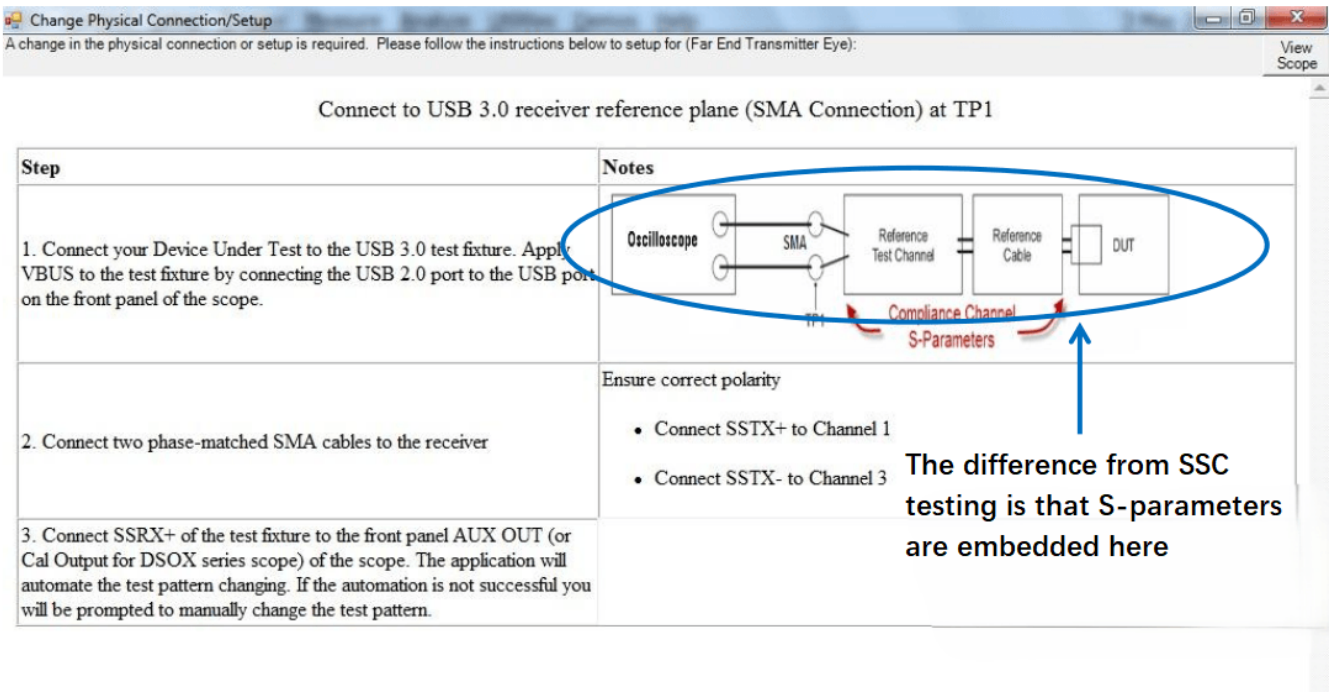


Figure 36 USB 3.0 Device Tx Eye Diagram and Jitter Test

5.4. Test completed, a test report is automatically generated. View the test report.

Margin Thresholds	
Warning	< 2 %
Critical	< 0 %

Pass	# Failed	# Trials	Test Name	Worst Actual	Worst Margin	Pass Limits
✓	0	2	5G LFPS Peak-Peak Differential Output Voltage	897.4 mV	24.4 %	800.0 mV <= VALUE <= 1.2000 V
✓	0	2	5G LFPS Period (tPeriod)	41.6532 ns	27.1 %	20.0000 ns <= VALUE <= 100.0000 ns
✓	0	2	5G LFPS Burst Width (tBurst)	939.4 ns	42.4 %	600.0 ns <= VALUE <= 1.4000 µs
✓	0	2	5G LFPS Repeat Time Interval (tRepeat)	10.0303 µs	49.6 %	6.0000 µs <= VALUE <= 14.0000 µs
✓	0	2	5G LFPS Rise Time	320.2 ps	92.0 %	VALUE <= 4.0000 ns
✓	0	2	5G LFPS Fall Time	326.4 ps	91.8 %	VALUE <= 4.0000 ns
✓	0	2	5G LFPS Duty cycle	51.3539 %	43.2 %	40.0000 % <= VALUE <= 60.0000 %
✓	0	2	5G LFPS AC Common Mode Voltage	42.6 mV	57.4 %	VALUE <= 100.0 mV
✓	0	1	5G TSSC-Freq-Dev-Min	-5.014726 kppm	17.8 %	-5.300000 kppm <= VALUE <= -3.700000 kppm
✓	0	1	5G TSSC-Freq-Dev-Max	25.001 ppm	45.8 %	TSSCMin ppm <= VALUE <= TSSCMax ppm
✓	0	1	5G SSC Modulation Rate	31.500990 kHz	50.0 %	30.000000 kHz <= VALUE <= 33.000000 kHz
✓	0	1	5G SSC Slew Rate	5.350 ms	46.5 %	VALUE <= 10.000 ms
✓	0	1	5G Short Channel Random Jitter	70 mUI	69.6 %	VALUE <= 230 mUI
✓	0	1	5G Short Channel Maximum Deterministic Jitter	221 mUI	48.6 %	VALUE <= 430 mUI
✓	0	1	5G Short Channel Total Jitter at BER-12	291 mUI	55.9 %	VALUE <= 660 mUI
✓	0	1	5G Short Channel Template Test	0.000	100.0 %	VALUE = 0.000
✓	0	1	5G Short Channel Differential Output Voltage	166.3 mV	6.0 %	100.0 mV <= VALUE <= 1.2000 V
✓	0	1	5G Far End Random Jitter (CTLE ON)	69 mUI	70.0 %	VALUE <= 230 mUI
✓	0	1	5G Far End Maximum Deterministic Jitter (CTLE ON)	211 mUI	50.9 %	VALUE <= 430 mUI
✓	0	1	5G Far End Total Jitter at BER-12 (CTLE ON)	280 mUI	57.6 %	VALUE <= 660 mUI
✓	0	1	5G Far End Template Test (CTLE ON)	0.000	100.0 %	VALUE = 0.000
✓	0	1	5G Far End Differential Output Voltage (CTLE ON)	113.7 mV	1.2 %	100.0 mV <= VALUE <= 1.2000 V

Figure 37 USB 3.0 Device Tx Test Report

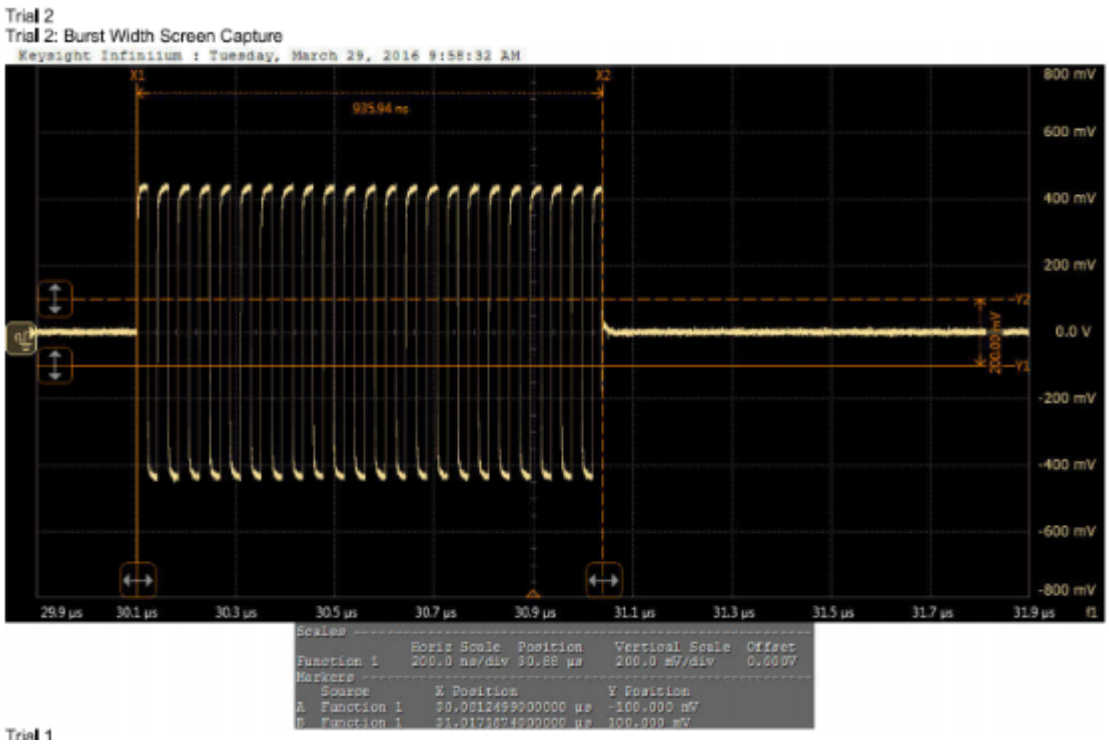


Figure 38 USB 3.0 Device Tx LFPS Burst Width

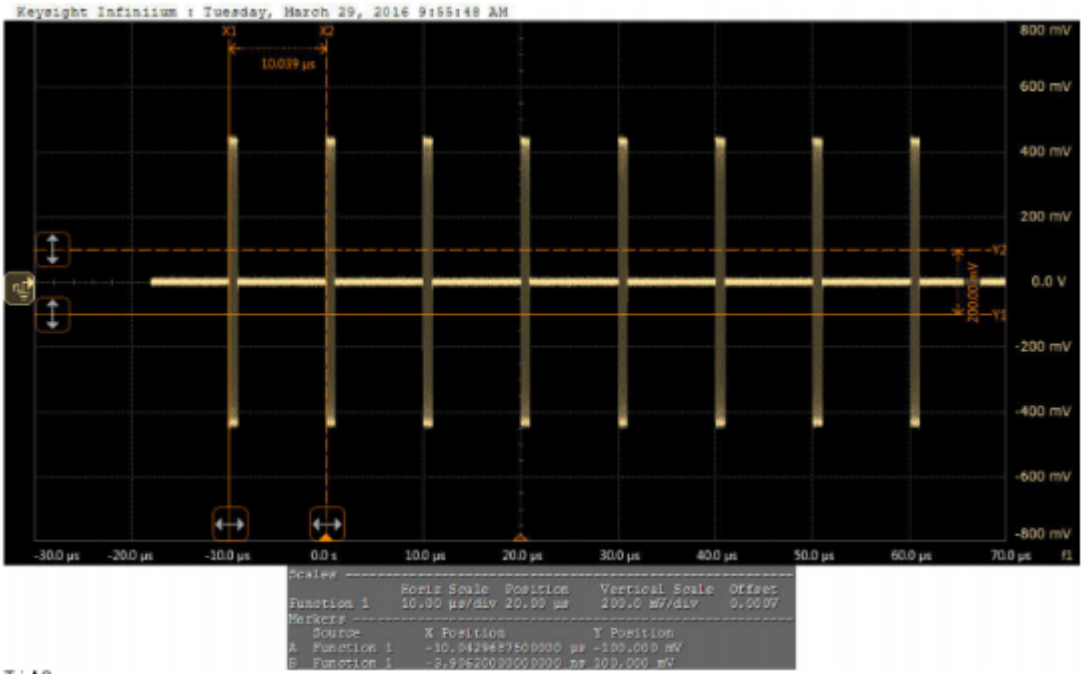


Figure 39 USB 3.0 Device Tx LFPS Repeat Time Interval

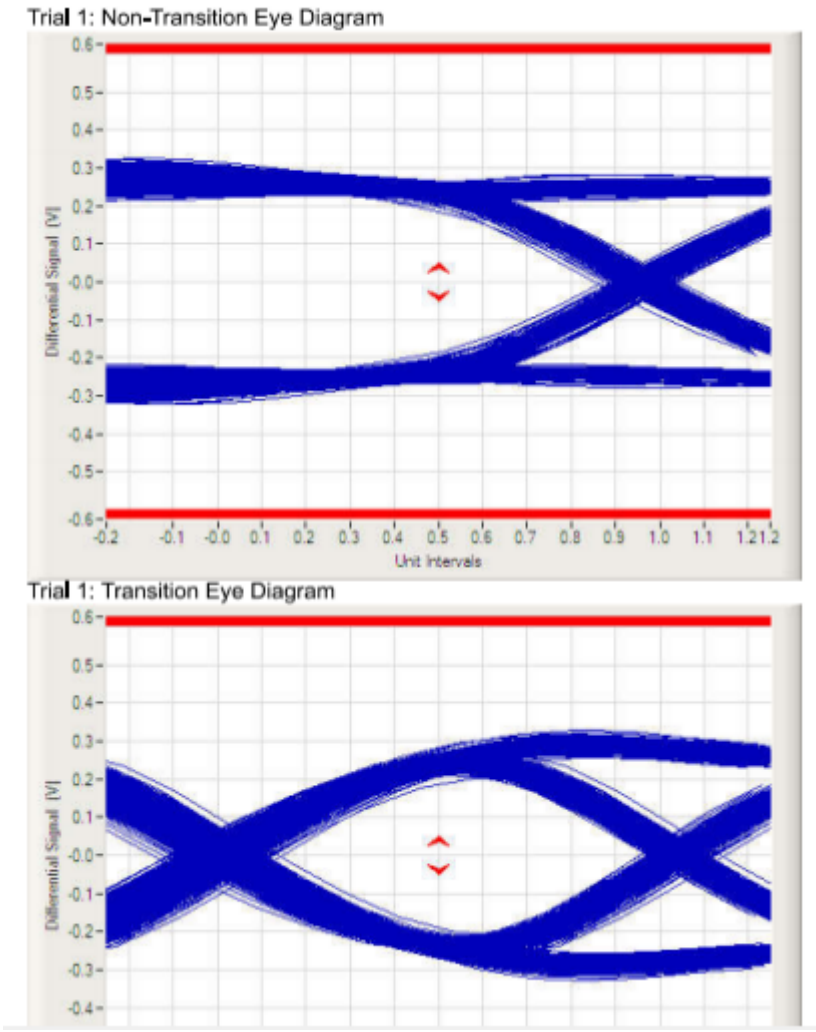


Figure 40 USB 3.0 Device Tx Short Channel Eye Diagram

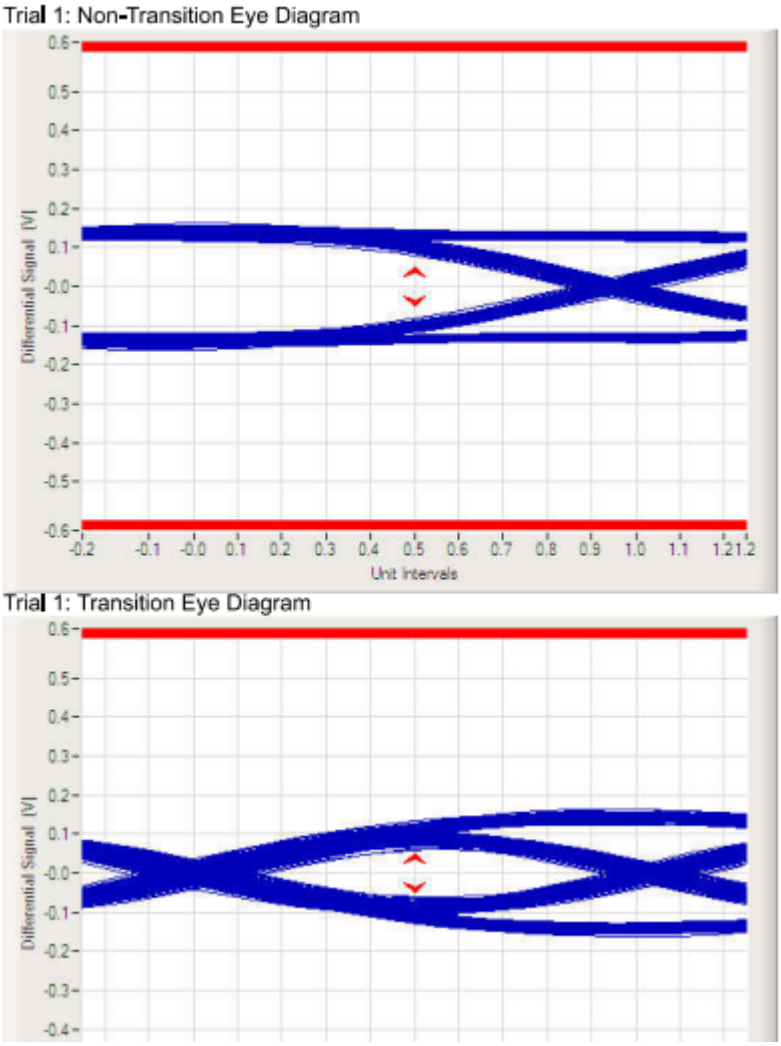


Figure 41 USB 3.0 Device Tx Far End Eye Diagram

3.2.7 USB 3.0 Host Tx Test Commands

Table 7 USB 3.0 Host Tx Test Commands

Chip Name	DWC3_0 Host USB 3.0	DWC3_1 Host USB 3.0	DWC3_2 Host USB 3.0
RK1808	io -4 0xff384008 0xc io -4 0xfd000430 0x0a010340	N.A	N.A
RK1820	io -4 0x1025000430 0x0a010340	N.A	N.A
RK3228H RK3328	io -4 0xff478408 0xc io -4 0xff600430 0x0a010340	N.A	N.A
RK3366	io -4 0xff500430 0x0a010340	N.A	N.A
RK3399	io -4 0xfe800430 0x0a010340	io -4 0xfe900430 0x0a010340	N.A
RK3528	io -4 0xfe500430 0x0a010340	N.A	N.A
RK3562	io -4 0xfe500430 0x0a010340	N.A	N.A
RK3566 RK3568	io -4 0xfcc00430 0x0a010340	io -4 0xfd000430 0x0a010340	N.A
RK3572	io -4 0x24000430 0x0a010340	io -4 0x24400430 0x0a010340	N.A
RK3576	io -4 0x23000430 0x0a010340	io -4 0x23400430 0x0a010340	N.A
RK3588 RK3588S	io -4 0xfc000430 0x0a010340	io -4 0xfc400430 0x0a010340	io -4 0xfcd00430 0x0a010340
RV1126B	io -4 0x21500430 0x0a010340	N.A	N.A

Note:

- *RK1808/RK3228H/RK3328 require two test commands to be executed, where the first command (i.e., writing "0xc") is to manually trigger the CP1 test pattern. Under normal circumstances, both commands can be input simultaneously during testing. However, if there is an abnormality in CP0 or CP1 test pattern switching, please do not execute the first command (i.e., writing "0xc") first. Instead, before starting the test, input the command to write the value "0x0a010340", and then during the test, when the oscilloscope prompts the CP1 pattern test window, input the command to write the value "0xc".*
- *The io command is applicable to all chips and kernel versions*
- *Linux-3.10/4.4 kernel versions also support testing RK3399 Type-C USB by writing to kernel device nodes. For the specific method, please refer to the section RK3399 USB 3.0 Test Issues*

3.2.8 USB 3.0 Host Tx Test Method

This document mainly describes the USB 3.0 Host Tx test method using the Agilent 90000 series oscilloscope (model: DSO91204A and test fixture U7242A). If you are using Tektronix or LeCroy oscilloscopes, please search for the test guides published officially by Tektronix and LeCroy.

3.2.8.1 USB 3.0 Host Tx Test Notes

- Before testing the USB 3.0 interface, it is recommended to first plug a USB 3.0-compatible peripheral (such as a USB 3.0 Disk) into the USB 3.0 interface to be tested, and query the enumeration information of the USB 3.0 peripheral through the test platform's debug serial console. If the enumeration information printed on the serial console includes "**SuperSpeed**", it indicates that the USB interface under test can properly support USB 3.0 functionality. Then unplug the USB 3.0 Disk and start testing. **If this step is omitted, the RK3399 Type-A USB 3.0 may not be able to enter test mode**;
- Before starting the test, you must first look up the test commands in Table 7 based on the chip platform and USB interface to be tested, and input the corresponding test commands before starting the test, in order to trigger the USB 3.0 controller to enter Host Tx test mode;
- The VBUS of the USB interface to be tested needs to output 5V power externally, while the test fixture U7242A does not need 5V power supply (this is the opposite of Device Tx testing);
- During the test procedure, **the timing of inputting test commands for Type-A and Type-C interfaces is different**. Please refer to the test step descriptions below for details;

3.2.8.2 USB 3.0 Host Tx Test Steps

1. Set up the automated test software. During USB 3.0 Host Tx testing, the setup of the oscilloscope's automated test software is similar to that of USB 3.0 Device Tx. Therefore, please refer to the description in USB 3.0 Device Tx Test Steps for the automated test software setup, which will not be repeated here;
2. First connect one end of the test fixture to the oscilloscope. **Do not connect the other end of the test fixture to the USB 3.0 Host interface of the device under test yet**;
3. Set the oscilloscope to the USB 3.0 LFPS test item. The oscilloscope will prompt to disconnect the test fixture from the USB 3.0 Host interface under test;
4. Based on the chip platform and USB interface to be tested, look up Table 7 USB 3.0 Host Tx Test Commands;
5. If the USB interface being tested is Type-A, first input the test command from step 4, then connect the test fixture to the USB 3.0 interface under test;

   If the USB interface being tested is Type-C, first connect the Type-C test fixture to the USB 3.0 interface under test, wait for the USB driver software to automatically switch to Host mode (wait 2~3 seconds), then input the test command from step 4;
6. Connect the test fixture to the USB 3.0 Host port to be tested, and the USB 3.0 controller will automatically enter test mode;
7. Follow the oscilloscope operation prompts to complete all test items;

3.2.9 USB 3.0 Tx Compliance Mode Query

- Device Tx Compliance mode query method

  `cat /sys/kernel/debug/usb/xxxx/link_state` ("xxxx" represents the node name of the USB controller to be tested, fill in according to the actual test platform)

  Return value: Compliance, indicating that Device Tx test mode has been entered

- Host Tx Compliance mode query method

  (1) Linux-4.19 and earlier kernel versions

  Read the PORTSC register of the USB 3.0 xHCI controller via the io command, bit[8:5] Port Link State (PLS). If PORTSC.PLS = 10 (decimal), it indicates that it is already in Compliance mode. For different chips, the base address of the PORTSC register may be different. Please check the chip TRM. The register offset address is fixed at 0x430.

  For example, the PORTSC register address of RK3399 USB 3.0 Host0 is 0xfe800430, and the PORTSC register address of USB 3.0 Host1 is 0xfe900430.

  (2) Linux-5.10 and newer kernel versions

  Linux-5.10 and newer kernel versions provide a portsc node, which allows for more convenient and intuitive querying of the Host Tx Compliance mode status. The node path is as follows:

  `cat /sys/kernel/debug/usb/xhci/xxxx/ports/port02/portsc` ("xxxx" represents the node name of the xHCI controller to be tested, fill in according to the actual test platform)

  Return value: Link:Compliance mode, indicating that Host Tx test mode has been entered

3.3 USB 3.0 Rx Compliance Test

The electrical performance test of USB 3.0 Rx is called Receiver Jitter Tolerance Test. During the test, **no test commands need to be input**. As long as the test environment is set up, the USB 3.0 controller will automatically enter **Loopback mode** after connecting to the test instrument and start Rx testing. Since the USB 3.0 Rx test environment setup is relatively complex, and the test steps differ for different oscilloscopes, this document does not provide detailed Rx test methods. Please refer to the operating instructions of the test oscilloscope.

This document only briefly explains the principle of entering Loopback mode and the method to confirm entry into Loopback mode.

3.3.1 USB 3.0 Rx Test Method

- Flow to enter Loopback mode

  During the Polling.Configuration phase of link training, if the USB 3.0 controller detects the Loopback bit in the T2 pattern sent by the test instrument, it will automatically configure the USB 3.0 PHY to enter Loopback mode. As shown in Figure 42 below.

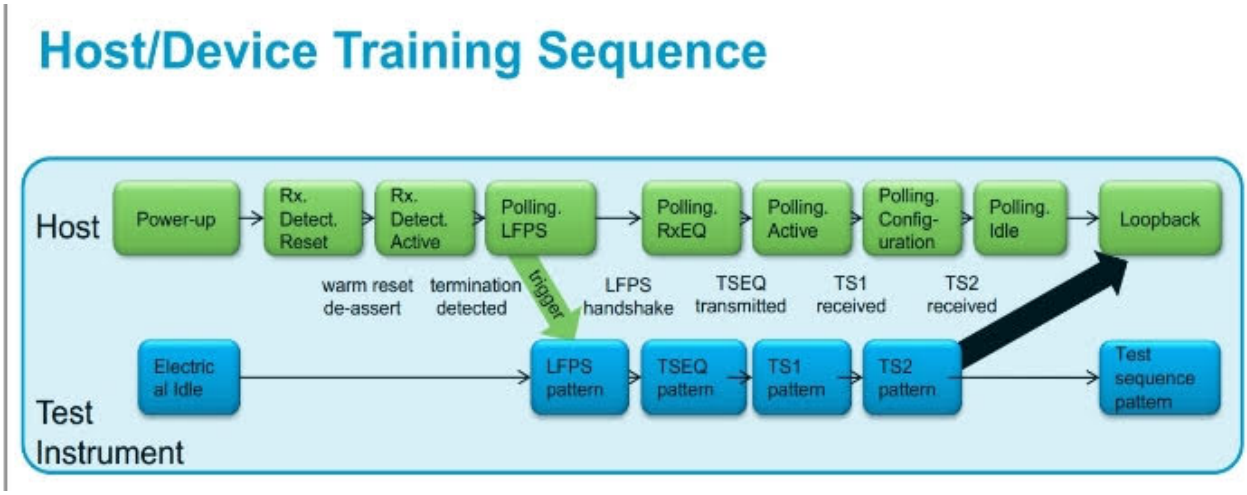


Figure 42 USB 3.0 Flow to Enter Loopback Mode

- Receiver Jitter Tolerance Test

  For the receiver jitter tolerance test method, please refer to the document Electrical Compliance Test Specification for SuperSpeed USB Rev. 1.0a

  The TD.1.5 Receiver Jitter Tolerance Test section describes the receiver jitter tolerance test steps in detail.

3.3.2 USB 3.0 Rx Loopback Mode Query

- Device Rx Loopback mode query method

  `cat /sys/kernel/debug/usb/xxxx/link_state` ("xxxx" represents the node name of the USB controller to be tested, fill in according to the actual test platform)

  Return value: Loopback, indicating that Device Rx test mode has been entered

- Host Rx Loopback mode query method

  (1) Linux-4.19 and earlier kernel versions

  Read the PORTSC register of the USB 3.0 xHCI controller via the io command, bit[8:5] Port Link State (PLS). If PORTSC.PLS = 11 (decimal), it indicates that it is already in Loopback mode. For different chips, the base address of the PORTSC register may be different. Please check the chip TRM. The register offset address is fixed at 0x430.

  For example, the PORTSC register address of RK3399 USB 3.0 Host0 is 0xfe800430, and the PORTSC register address of USB 3.0 Host1 is 0xfe900430.

  (2) Linux-5.10 and newer kernel versions

  Linux-5.10 and newer kernel versions provide a portsc node, which allows for more convenient and intuitive querying of the Host Rx Loopback mode status. The node path is as follows:

  `cat /sys/kernel/debug/usb/xhci/xxxx/ports/port02/portsc` ("xxxx" represents the node name of the xHCI controller to be tested, fill in according to the actual test platform)

  Return value: Link:Test mode, indicating that Host Rx test mode has been entered

3.4 USB 3.0 Compliance Test Common Issues and Solutions

3.4.1 USB 3.0 Device Tx Unable to Enter Test Mode

Under normal circumstances, USB 3.0 Device does not need to input test commands and can automatically enter Tx test mode. If it cannot enter test mode, it is recommended to test the VBUS voltage first. According to the USB 3.0 Device Tx Compliance test specification, the **VBUS 5V of USB 3.0 Device must not be self-powered**, otherwise the USB 3.0 controller will not be able to automatically enter Compliance mode. The VBUS power supply must be provided by the test fixture. You can connect the USB power port of the test fixture to the USB port of the oscilloscope or PC via a USB cable to provide VBUS 5V power.

3.4.2 RK3399 USB 3.0 Test Issues

1. RK3399 Type-A USB 3.0 Host unable to enter Tx test mode

   When testing RK3399 Type-A USB 3.0 Host, an additional step is required: before testing, first plug a USB 3.0 Disk into the USB 3.0 interface to be tested. For detailed operations, see USB 3.0 Host Tx Test Method —— USB 3.0 Host Tx Test Notes. Otherwise, the Type-A USB 3.0 may not be able to enter test mode.

   In addition to the above method, this can also be resolved by modifying the software. The specific method is to delete the following property from the usbdrd_dwc3_0 and usbdrd_dwc3_1 nodes in the DTS:

   `snps,usb3-warm-reset-on-resume-quirk`

2. RK3399 Type-C USB 3.0 Host unable to enter Tx test mode

   Under normal circumstances, RK3399 Type-C USB 3.0 Host can enter Tx test mode by simply referring to USB 3.0 Host Tx Test Method. If it cannot enter Tx test mode, it is recommended to test the VBUS voltage first. If VBUS is constantly powered, it may prevent entering Tx test mode. There are two solutions:

   Method 1: Configure VBUS power supply as software-controllable, i.e., it does not output 5V by default and only outputs 5V when a Type-C to Type-A cable is inserted;

   Method 2: Configure the corresponding dr_mode property in the DTS as "host";

3. RK3399 Type-A USB 3.0 Host unable to enter Rx Loopback mode

   If testing the **Type-A USB 3.0** interface, it is recommended to first plug a USB 3.0 Disk into the USB 3.0 interface to be tested before starting the test, and confirm via serial console log that it can be recognized as "SuperSpeed", indicating that the test firmware can properly support USB 3.0. Then unplug the USB 3.0 Disk and start testing. If this step is omitted, RK3399 Type-A USB 3.0 may not be able to enter Loopback mode.

4. USB 3.0 Host Tx test method for Linux-3.10/4.4 kernels on RK3399 platform

   Note: Only applicable to RK3399 chips

   1. Linux-3.10/early Linux-4.4 versions on RK3399 platform

      Type-C0 USB front side:

      ```
      echo test_u3 > /sys/kernel/debug/usb@fe800000/host_testmode
      ```

      Type-C0 USB reverse side:

      ```
      echo test_flip_u3 > /sys/kernel/debug/usb@fe800000/host_testmode
      ```

Type-C1 USB front side:

`echo test_u3 > /sys/kernel/debug/usb@fe900000/host_testmode`

Type-C1 USB reverse side:

`echo test_flip_u3 > /sys/kernel/debug/usb@fe900000/host_testmode`

2. Latest Linux-4.4 version on RK3399 platform

   Type-C0 USB front side:

   `echo test_u3 > /sys/devices/platform/usb0/host_testmode`

   Type-C0 USB reverse side:

   `echo test_flip_u3 > /sys/devices/platform/usb0/host_testmode`

   Type-C1 USB front side:

   `echo test_u3 > /sys/devices/platform/usb1/host_testmode`

   Type-C1 USB reverse side:

   `echo test_flip_u3 > /sys/devices/platform/usb1/host_testmode`

You can execute the following command to check whether the USB has entered test mode:

`cat /sys/kernel/debug/xxxx/host_testmode` ("xxxx" represents the usb3 controller node name to be tested, fill in according to the actual test platform)

The returned results are as follows:

U2: test_packet // means that U2 is in test mode

U3: compliance mode // means that U3 is in test mode

(If U3: UNKNOWN is returned, it means the USB has not entered test mode)

3.4.3 RK1808 USB 3.0 Test Issues

1. RK1808 Type-A USB 3.0 Host unable to enter Rx Loopback mode

   By modifying the software to disable the hub autosuspend function, the modification reference is as follows:

```
diff --git a/drivers/usb/core/usb.c b/drivers/usb/core/usb.c
index 36e5098..0b2930d 100644
--- a/drivers/usb/core/usb.c
+++ b/drivers/usb/core/usb.c
@@ -66,7 +66,7 @@ int usb_disabled(void)
 EXPORT_SYMBOL_GPL(usb_disabled);

 #ifdef CONFIG_PM
-static int usb_autosuspend_delay = 2;          /* Default delay value,
+static int usb_autosuspend_delay = -1;         /* Default delay value,
```

3.4.4 RK3588 USB 3.0 Test Issues

RK3588 USB 3.0 Tx/Rx testing is based on the Linux-5.10 kernel. Notes are as follows:

1. When testing the USB 3.0 Device mode of the Type-C0 interface, you need to first disable the controller's autosuspend function. The command is as follows:

   `echo on > /sys/devices/platform/usbdrd3_0/fc000000.usb/power/control`

2. When testing the USB 3.0 Host mode of the Type-C interface, you need to first connect the Type-C test fixture to the USB 3.0 interface under test, wait for the USB driver software to automatically switch to Host mode (wait 2~3 seconds), and then input the Host test command from Table 7.

3. When testing USB 3.0 Rx functionality, if it cannot automatically enter loopback mode, you need to modify the DWC3 controller configuration in the DTS to add:

   `snps,dis_u3_susphy_quirk;`

4. Method to confirm that Type-C0 USB 3.0 Device mode Tx has entered test mode

   `cat /sys/kernel/debug/usb/fc000000.usb/link_state`

   Return value: Compliance, indicating that Device Tx test mode has been entered

5. Method to confirm that Type-C0 USB 3.0 Device mode Rx has entered test mode

   `cat /sys/kernel/debug/usb/fc000000.usb/link_state`

   Return value: Loopback, indicating that Device Rx test mode has been entered

6. Method to confirm that USB 3.0 Host mode Tx has entered test mode

   Taking Type-C1 USB 3.0 Host as an example:

   `cat /sys/kernel/debug/usb/xhci/xhci-hcd.8.auto/ports/port02/portsc`

   Return value: Link:Compliance mode, indicating that Host Tx test mode has been entered

7. Method to confirm that USB 3.0 Host mode Rx has entered test mode

   Taking Type-C1 USB 3.0 Host as an example:

   `cat /sys/kernel/debug/usb/xhci/xhci-hcd.8.auto/ports/port02/portsc`

   Return value: Link:Test mode, indicating that Host Rx test mode has been entered

4. USB 3.0 HUB Compliance Test

The Compliance test of USB 3.0 HUB includes upstream ports and downstream ports. However, in practical applications, we generally only need to test the downstream ports provided for user use. Therefore, this document only provides the Compliance test method for testing USB 3.0 HUB downstream ports.

Common USB 3.0 HUB chip models mainly include: GL352x series, VL812, VL813, USB5734, RTS5411, CYPRESS HX3 series, etc. Unlike the USB 2.0 HUB test method, the USB 3.0 HUB Compliance Test on the Rockchip platform can only use the script test method.

1. Download and compile the test script:

   Script source code download link: https://redmine.rockchip.com.cn/documents/113

   File: USB_HUB_Compliance_Test_Script

2. Execute the test script:

Taking the RK3399 platform testing GL3523 HUB as an example, the test steps are as follows:

(1) Use adb push to push the script to the Android system

`adb push C:\Users\user\Desktop\linux-eye /data`

(2) Modify the permissions of linux-eye

`root@rk3399:/data # chmod 777 linux-eye`

(3) Execute the script to set the USB3 HUB port to test mode

(3.1) Determine the USB 3.0 HUB information to be tested based on kernel log

```
[  139.427845] usb 6-1: new SuperSpeed USB device number 2 using xhci-hcd
[  139.445641] usb 6-1: New USB device found, idVendor=05e3, idProduct=0612
[  139.445708] usb 6-1: New USB device strings: Mfr=1, Product=2, SerialNumber=0
[  139.445738] usb 6-1: Product: USB 3.0 Hub
[  139.445763] usb 6-1: Manufacturer: GenesysLogic
[  139.452409] usb 5-1: new high-speed USB device number 2 using xhci-hcd
[  139.463572] hub 6-1:1.0: USB hub found
[  139.465861] hub 6-1:1.0: 4 ports detected
[  139.589854] usb 5-1: New USB device found, idVendor=05e3, idProduct=0610
[  139.589920] usb 5-1: New USB device strings: Mfr=1, Product=2, SerialNumber=0
[  139.589950] usb 5-1: Product: USB 2.0 Hub
[  139.589975] usb 5-1: Manufacturer: GenesysLogic
[  139.607244] hub 5-1:1.0: USB hub found
[  139.609146] hub 5-1:1.0: 4 ports detected
```

(3.2) Execute the test script

```
root@rk3399:/ # ./data/linux-eye

LinuxEye - select one of the following hub for testing.

        [ 0]    4-port Super-Speed hub at tier 2 of Bus 6
                (VID: 05E3, PID: 0612, Address: 2)

        [ 1]    4-port High-Speed hub at tier 2 of Bus 5
                (VID: 05E3, PID: 0610, Address: 2)

    Please enter [0 ~ 1] to select a hub or 'q' to quit: 0 (Enter 0, indicating testing
super-speed)
                [ 1] is open
                [ 2] is open
                [ 3] is open
                [ 4] is open
    Please enter [1 ~ 4] to select a port or 'q' to quit: 1 (Enter 1, indicating
testing USB3 HUB port1. To test port2, enter 2, and so on)
    device file /dev/bus/usb/006/002 opened successfully
           Port (1) Status: 02A0
    LinuxEye - Start testing port 1 of device 2 on bus 6    (Start testing)
               Type 'q' to stop the test: q                 (Test finished, enter q to
quit)
```

(3.3) Repeat the above steps to test other USB 3.0 HUB downstream ports

5. Rockchip USB SQ Tool

Rockchip USB SQ Tool is a USB PHY signal integrity debugging tool self-developed by Rockchip. It can support querying USB signal test commands and dynamically adjusting USB PHY signals, and can also automatically generate the corresponding PHY driver code based on the PHY Tuning results. For detailed instructions on USB SQ Tool, please refer to the document "Rockchip_Introduction_USB_SQ_Tool_EN".

USB SQ Tool download link: https://redmine.rockchip.com.cn/documents/113

6. Reference Documents

1. USB 2.0 Specification
2. USB 3.1 Specification
3. Agilent N5416A USB 2.0 Compliance Test Option
4. Agilent USB 2.0 High Speed Device SQ Test
5. Keysight N7015A-16A Type-C Test Kit
6. USB 2.0 Electrical Compliance Test Specification Version 1.07
7. Electrical Compliance Test Specification for SuperSpeed USB Rev. 1.0a